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SCHOOL OF BUSINESS, ECONOMICS AND LAW

Does Drug Repurposing Lead to Faster FDA Approval? Evidence from New Drug Applications, 1990–1999

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Master's thesis in Innovation & Industrial Management

Spring 2026

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Declaration of AI and Tool Usage

This thesis has been supported by several digital tools, which are hereby disclosed as per the policy of the School of Business, Economics, and Law at the University of Gothenburg.

Claude (Anthropic) was the main digital tool for this project at different times and for quite a few very specific tasks. At early stages, it was the main tool for finding, understanding, and structuring the research topic, as well as for mapping the theoretical frameworks and identifying the relationships among the key concepts. During the writing phase, it was like a tutor and a partner in thinking, helping me understand the methodological decisions that went into the analysis, such as why a negative binomial regression model was used and how to read statistical outputs like incidence rate ratios. It was also of great help in making my points and arguments clearer, especially in the discussion and findings chapters, when I was focused on expressing my interpretations accurately. Moreover, Claude worked alongside me when I wrote the R code to do the empirical analysis and generate the figures and tables presented in this thesis.

Grammarly (Free Version) was used to check sentence structure, grammar, and readability across the thesis document.

All research questions, hypotheses, analytical decisions, arguments, and conclusions submitted within this thesis are my original work without any assistance or contribution from others. I bear full responsibility for the content, accuracy, and integrity of my final work. The above-described tools served as an aid and stimulation to my thinking and writing; however, they did not replace my original work. I have thoroughly read, understood, and critically evaluated all the references mentioned in this thesis.

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Acknowledgement

I express my deep gratitude to my supervisor, Dr Bastian Marius Rake, for his vital guidance and support throughout this research. Beyond providing the dataset that formed the foundation of this study, Dr Rake offered detailed and constructive feedback at every stage of the process, from refining the theoretical framework and strengthening the hypothesis development to recommending the appropriate statistical methods for the analysis. His suggestion to use negative binomial regression as the primary model, and his encouragement to engage more deeply with the exaptation and bricolage literature, significantly improved the quality and rigour of this thesis. I am grateful for both his academic expertise and his patience in responding to my questions throughout the process.

I also thank my wife, Puja, whose steady support made this work possible. Completing a Master's while managing daily life is not easy, and her dedication, patience, and constant effort kept me focused during the hardest moments. This thesis reflects her sacrifice and hard work as much as my own, and I am truly grateful to have had her by my side on this journey.



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Gothenburg

31 May 2026

List of Abbreviations

Acronym	Full Form
CI	Confidence Interval
FDA	Food and Drug Administration
H1	Hypothesis 1
H2	Hypothesis 2
IRR	Incidence Rate Ratio
NB	Negative Binomial
NDA	New Drug Application
NME	New Molecular Entity
OLS	Ordinary Least Squares
OTC	Over the Counter
PDUFA	Prescription Drug User Fee Act
R&D	Research and Development
SD	Standard Deviation
SE	Standard Error
US	United States
USD	United States Dollar

Abstract

Drug repurposing has gained interest as a faster, cheaper path than traditional drug development, with industry claims of up to 85% lower costs and five years shorter timelines. Yet direct evidence that repurposed drugs obtain regulatory approval faster is scarce. This study fills that gap by comparing approval times for repurposed versus non-repurposed drugs using FDA Drugs@FDA data. This study examines 289 Type 1 new molecular entity NDA submissions from 1990 to 1999, of which 148 had full approval-duration records. The main analysis uses negative binomial regression, with OLS on log-transformed duration as a robustness check, and controls for review priority, submission year, and marketing status.

Repurposed drugs averaged 538 days from submission to approval, while non-repurposed drugs averaged 609 days, a 71-day advantage for repurposing. However, bivariate tests and regression models did not find this difference statistically significant. The key finding is institutional: submission year, reflecting the effect of the Prescription Drug User Fee Act of 1992, is the strongest predictor of approval time, with each later year linked to about an 18% shorter duration. After accounting for this time trend, the repurposing status adds little explanatory power. The study concludes that any theoretical time savings from repurposing do not clearly appear at the regulatory approval stage in this sample, implying that efficiency gains likely occur earlier in development before submission.

Keywords: drug repurposing, FDA approval, regulatory timelines, negative binomial regression, PDUFA, pharmaceutical innovation, exaptation, R&D productivity

Chapter 1: Introduction

The pharmaceutical industry is one of the most research-intensive sectors in the global economy. However, it has been operating under a productivity crisis that has been intensifying since the 1950s, whose time to some extent corresponds to the inception of the industry's growth. The productivity of the sector, measured in terms of the number of drug approvals per billion dollars spent, has kept on falling continuously during this period. Scannell et al. (2012) named this trend Eroom's Law by flipping Moore's Law of computing, which traced the improvement of semiconductor production efficiency, and stated that the productivity of pharmaceutical R&D has been declining approximately eighty times on a constant dollar basis during the period. The reasons are structural rather than incidental. Paul et al. (2010) reveal that development timelines starting from discovery to approval in the market take on average more than a decade; according to DiMasi et al. (2016), the average cost of a new drug is over USD 2.5 billion, and Hay et al. (2014) indicate that of those compounds that enter Phase I clinical trials, regulatory approval is granted only to about 10%. Taking these figures together, they paint a picture of an industry in serious and sustained pressure.

It is worth acknowledging the picture isn't entirely bleak. For example, FDA approval numbers have gone up in the more recent years, and some reform efforts from the industry have been truly impactful. But the fundamental structural problems have not been fixed. For example, rising development costs, maintaining high attrition rates in clinical trials, and no significant shortening of timelines are still unresolved. Moreover, the need for discovering more efficient approaches of producing therapeutic and commercial value has not only remained but also increased.

In this backdrop, drug repurposing has been rising to become one of the most strategically important alternative methods to pharmaceutical innovation. Repurposing, or repositioning, as it is also called, is finding and developing new therapeutic applications for drugs that have already been tested and approved for use in different indications. Since a repurposed drug already has a known safety profile and pharmacological properties, the initial development phases can sometimes be shortened or avoided completely, thereby saving both time and money when compared to a new compound developed from scratch (Ashburn and Thor, 2004; Pushpakom et al., 2019). According to industry estimates, repurposing can cut development costs by 85% and reduce development times by five years (Paul et al., 2010; Rake and Hanisch, 2023). According to Rake and Hanisch (2023), repurposing efforts may actually be responsible

for as much as 30% of FDA-approved drugs, thereby making it a regular part of pharmaceutical innovation rather than an offbeat strategy.

The examples of sildenafil and thalidomide probably serve as the most well-known cases that illustrate how repurposing operates in reality. First, sildenafil, which was initially intended by Pfizer for angina, was turned into what we now know as Viagra after various unintentional clinical observations. And thalidomide, after causing birth defects at a massive scale, was withdrawn at the beginning of the 1960s. Many years later, after it was discovered that it works by modifying the immune system, it was finally approved for use in multiple myeloma. The two cases above are classic examples of how the old compounds, when their functions are recognised differently, e.g., in a new therapeutic setting, can lead to new treatment possibilities.

Repurposing is best understood not as a replacement for conventional drug development but as a complementary innovation strategy. It does not directly solve the productivity crisis by creating new molecular entities. What it does is allow firms to generate additional value from existing knowledge, serve commercially unattractive markets such as rare diseases, and do so at a fraction of the cost and risk of de novo development (Rake and Hanisch, 2023). These advantages are real and well established in the theoretical literature. However, one specific and frequently cited advantage, that repurposed drugs move through regulatory approval more quickly than non-repurposed drugs, has received surprisingly little direct empirical scrutiny.

This is not a trivial gap. As Breckenridge and Jacob (2019) point out, existing regulatory frameworks were not designed with repurposing in mind. The FDA evaluates each new indication on its own merits and may require extensive new clinical evidence regardless of how well established the compound's safety profile already is. Whether the theoretical time advantage of repurposing materialises at the regulatory approval stage is therefore an open empirical question, not something that follows automatically from first principles.

This thesis addresses that question directly. Using data from the FDA Drugs@FDA database, it compares regulatory approval durations between 122 repurposed and 167 non-repurposed drugs across 289 original Type 1 new molecular entity NDA submissions from 1990 to 1999. Approval durations were calculated in days using exact submission and approval dates extracted from original FDA approval letters. Repurposing status was determined by reviewing FDA approval histories for evidence of subsequent efficacy supplements, indicating approval for a new therapeutic indication. The analysis controls for review priority designation,

submission year, and marketing status, using negative binomial regression as the primary model and OLS on log-transformed duration as a robustness check.

The 1990 to 1999 period was chosen deliberately. It includes the Prescription Drug User Fee Act of 1992, which reformed FDA review timelines substantially and created meaningful institutional variation within the sample period. It also predates the widespread adoption of computational drug discovery, ensuring all drugs in the sample were developed under broadly comparable conditions. The complete documentation of all approvals in this period makes it well suited to a systematic empirical analysis.

This research makes a twofold contribution. First, it offers direct quantitative support for one aspect of repurposing efficiency that most of the prior research has only theorised but not empirically explored. Second, it takes the idea of repurposing in pharmaceuticals and links it to the wider innovation management literature on exaptation and bricolage, placing it within a theoretical framework that is not limited to pharmaceuticals only.

The remainder of the thesis is structured as follows. Chapter 2 reviews the existing literature on pharmaceutical R&D productivity and drug repurposing, develops the theoretical framework drawing on exaptation and bricolage, and derives the two hypotheses tested in this study. Chapter 3 describes the data, variables, and analytical strategy. Chapter 4 presents the empirical results. Chapter 5 discusses the findings in relation to the hypotheses and the existing literature. Chapter 6 concludes.

Chapter 2: Literature Review and Theoretical Framework

The pharmaceutical industry is one of the most research-intensive sectors in the world, but it has been struggling with a fundamental problem for a very long time: companies are investing more in developing new drugs, but fewer drugs are coming to the market. This is not just an academic observation. It has real consequences for patients who need new treatments, for companies trying to remain competitive, and for the broader question of how innovation in healthcare works. This chapter reviews the existing literature on this problem and on drug repurposing as one potential response to it. It also develops the theoretical framework that guides the empirical analysis and explains the research gap this thesis seeks to fill.

2.1 The Pharmaceutical R&D Productivity Problem

To understand why drug repurposing has attracted so much attention, it helps to first understand how difficult traditional drug development has become. The process of introducing a new drug to market involves multiple stages: early laboratory research, preclinical testing in animals, and then three phases of clinical trials in humans, followed by regulatory review. Each of these stages is expensive, time-consuming, and uncertain. And the problem is that over time, this process has become increasingly inefficient despite significant advances in science and technology.

Scannell et al. (2012) gave this problem a name: Eroom's Law. It is a deliberate inversion of Moore's Law, the observation from computing that processing power doubles roughly every two years. Eroom's Law says the opposite: the number of new drugs approved per billion dollars spent on R&D has halved approximately every nine years since 1950, falling around eighty times in inflation-adjusted terms. In other words, we are spending more and getting less. Scannell et al. (2012) argue that the underlying causes have not been properly identified, which is part of why proposed solutions have had limited impact.

The numbers behind this problem are striking. Paul et al. (2010) document that it takes more than a decade on average for a drug to move from initial discovery to market approval. Hay et al. (2014) show that only around 10% of compounds entering Phase I clinical trials ever make it through to approval, with particularly high failure rates at Phase II, where efficacy is tested most rigorously. DiMasi et al. (2016) estimate the average cost of developing a new drug at over 2.5 billion USD, once you account for the cost of all the failed projects that had to be funded alongside the successful ones.

What is important to understand about these three sets of findings is that they are connected, not separate issues. The high failure rate drives up costs because every failed project is a forgone cost as it consumes resources. The long timelines compound costs further because capital tied up in development cannot generate returns. Paul et al. (2010) frame this through a productivity model where R&D productivity depends on success rates, development time, and cost simultaneously. Improving any one of these dimensions would improve overall productivity, but the three are interlinked.

This situation has created real pressure on pharmaceutical companies to find alternative ways of innovating. Simply investing more money in traditional drug discovery has not worked. Something about the model itself needs to change. This is the context in which drug

repurposing has emerged as a serious strategic option, and it is also the context that gives the research question of this thesis its practical relevance.

2.2 Drug Repurposing as a Response to the Productivity Problem

Drug repurposing, which is also called drug repositioning or reprofiling, means finding new therapeutic uses for drugs that have already been developed and tested for something else. The core idea is straightforward: if a drug has already gone through safety testing for one disease, you already know a lot about how it behaves in the human body. That existing knowledge can potentially be used to shortcut the development process for a new indication (Ashburn and Thor, 2004; Pushpakom et al., 2019).

In the broader innovation management literature, repurposing is understood more generally as the transfer of a known solution to a new context, where the new application was not originally intended (Rake and Hanisch, 2023; 2024). This broader definition is useful because it connects pharmaceutical repurposing to a much wider phenomenon in innovation. The same basic concept applies whether you are repurposing a drug, a piece of software, or a manufacturing technology. You are leveraging something that already exists rather than building something entirely from scratch.

The scale of repurposing in the pharmaceutical industry is significant. Rake and Hanisch (2023) note that up to 30% of FDA-approved drugs are the result of repurposing efforts. During the COVID-19 pandemic, the proportion was even higher: Hanisch and Rake (2021, as cited in Rake and Hanisch, 2023) found that 86% of drugs tested against COVID-19 through July 2020 were repurposed compounds. These numbers suggest that repurposing is not a niche strategy but a mainstream operational feature of the pharmaceutical industry.

The efficiency argument for repurposing rests primarily on the idea of prior knowledge. When a drug has already passed through preclinical testing and early clinical trials for a previous indication, the company developing it for a new use does not need to start from zero. The safety profile is already established, the pharmacological properties are well known, and some of the regulatory groundwork has already been done. This means that early development stages, particularly Phase I trials focused on safety, can often be shortened or skipped entirely (Pushpakom et al., 2019). Parvathaneni et al. (2019) make a similar point, noting that repurposed drugs can enter clinical development at a more advanced stage than genuinely new compounds. Industry estimates suggest repurposing can reduce development costs by up to

85% and cut the development cycle by up to five years (Paul et al., 2010; Rake and Hanisch, 2023).

Repurposing also offers potential benefits beyond pure efficiency. Because the development costs are lower, it becomes economically viable to develop treatments for diseases that would otherwise be ignored because the market is too small to justify the investment. Rare diseases are the most obvious example. Rake and Hanisch (2023) note that repurposing can make otherwise unattractive niche markets commercially viable, which has implications for addressing unmet healthcare needs that conventional development models have failed to serve.

The history of repurposing includes some well-known cases. Sildenafil was originally developed for angina, but during clinical trials, researchers noticed an unexpected side effect that led to its repurposing as a treatment for erectile dysfunction (Pushpakom et al., 2019; Rake and Hanisch, 2023). Thalidomide was withdrawn in the 1960s after it was found to cause severe birth defects in children born to women who took it during pregnancy. Later, its immunomodulatory properties were recognised, and it was approved for multiple myeloma. These examples are frequently cited in the literature, and for good reason: they show clearly how unexpected properties of existing compounds can become the basis for entirely new therapeutic applications.

However, it would be a mistake to present repurposing as a proven solution to the productivity problem. There are real challenges. The most significant is intellectual property: because the compound itself is typically no longer patentable, companies relying on repurposing face difficulties protecting their investment (Ashburn and Thor, 2004; Breckenridge and Jacob, 2019). If a competitor can copy a repurposed drug relatively easily once it has been approved, the financial incentive to invest in repurposing in the first place is reduced. Breckenridge and Jacob (2019) specifically discuss the legal and regulatory barriers to repurposing, noting that existing frameworks were not designed with repurposing in mind and can create obstacles rather than facilitating it.

The other major challenge is regulatory complexity, and this one is directly relevant to the research question of this thesis. Even when a drug has established safety data, the FDA may still require extensive new clinical trials depending on how different the proposed new indication is from the original one. This means that the time advantage attributed to repurposing is not automatic. Whether repurposed drugs move through the regulatory approval process faster than non-repurposed drugs is an empirical question, not something that can simply be

assumed from the theoretical arguments. This observation is central to the motivation for the present study.

Interestingly, the efficiency-related claims of the repurposing literature are mainly about the pre-submission stages of drug development. These efficiency claims focus on the shortening of the preclinical testing as well as Phase I trials due to the availability of the safety data of the compound in question. The question is whether these time savings can also be achieved at the FDA regulatory review stage, which is, actually, what this study has set out to investigate.

2.3 Theoretical Foundations: Understanding How Repurposing Works

The efficiency arguments reviewed in the previous section explain why repurposing might be faster and cheaper than conventional development, but they do not fully explain the underlying mechanism. To understand repurposing as an innovation strategy more deeply, it helps to draw on theoretical concepts from the innovation management literature. Two concepts are relevant: bricolage and exaptation. These ideas come from different intellectual traditions, but they converge on a similar insight: that significant innovation can emerge from the creative reuse of existing resources and knowledge, rather than from the development of entirely new ones.

2.3.1 Bricolage

The concept of bricolage comes originally from the work of the anthropologist Lévi-Strauss and was developed for the study of entrepreneurship and innovation by Baker and Nelson (2005). At its core, bricolage describes improvised problem-solving using whatever resources are immediately available, rather than acquiring purpose-built solutions. The classic illustration offered by Rake and Hanisch (2023) is the Apollo 13 emergency, where NASA engineers had to construct a CO₂ filter from plastic bags, duct tape, and other materials that happened to be on the spacecraft. It worked, but it was clearly a makeshift solution rather than a systematic approach to innovation.

In the context of pharmaceutical repurposing, bricolage describes the historical pattern of serendipitous discovery. Many early repurposing successes, including sildenafil and thalidomide, came about because clinicians or researchers noticed unexpected effects in drugs they were already working with. They used what was available rather than setting out to find a solution. This kind of discovery is valuable when it happens, but it is not something that can be planned or systematically managed. Rake and Hanisch (2023) make the point that bricolage-based repurposing, while sometimes productive, typically produces solutions that are limited

in performance and cannot be relied upon as a core innovation strategy. This former study here shows how, theoretically, bricolage is opposed to exaptation. The main idea behind bricolage is that it refers to the historical way in which the lack of discovery has been repurposed by random means, which cannot be planned systematically. In contrast, exaptation is a planned and certified formal method which is the one that is used in modern repurposing strategy and on which efficiency arguments of this thesis are tested empirically.

2.3.2 Exaptation

'Exaptation' offers a more theoretically powerful concept of how systematic repurposing works. The concept originated from evolutionary biology, where it describes how a biological feature that evolved for one purpose gets co-opted for a completely different function. The most frequently cited example is feathers, which appear to have evolved originally for thermal regulation but were later adapted for flight. The key point is that the new function was not what drove the original development; it emerged from properties that were already present but previously incidental.

Andriani and colleagues have done the most to develop exaptation as a concept for understanding innovation. Andriani et al. (2017) argue that exaptation represents a distinct mode of search and problem-solving in which novelty emerges from the redeployment of existing solutions rather than from dedicated efforts to solve a specific problem. Their empirical work demonstrates that exaptive innovations are systematically different from conventional innovations in how they are discovered and how they perform. Andriani and Kaminska (2021) extend this through a historical analysis of innovations in the coal tar industry, showing how a series of major innovations, including pharmaceuticals, dyes, and synthetic materials, emerged through exaptive redeployment of existing chemical knowledge.

Mastrogiorgio and Gilsing (2016) contribute an important further insight: exaptation is not random. It is more likely under certain conditions, particularly when the original technology is complex enough to contain multiple latent properties that were not fully exploited in its original application and when the search strategy is broad enough to recognise potential applications across different domains. This has practical implications for drug repurposing: compounds with complex mechanisms of action that affect multiple biological pathways may be more likely repurposing candidates precisely because their pharmacological profiles contain more latent potential.

In the pharmaceutical context, exaptation maps directly onto the logic of repurposing. A compound developed and tested for one indication has properties that were characterised during its original development. Some of those properties may turn out to be relevant for an entirely different disease. The transition from sildenafil as an angina treatment to sildenafil as an erectile dysfunction treatment is a textbook case of exaptation: the vasodilatory properties of the drug, which were known from the original development, turned out to have applications in a context that was not the original focus of research.

The theoretical significance of exaptation for this thesis is that it provides a mechanism-level explanation for why repurposing might be faster than conventional development. If a repurposed drug is being deployed for a new indication based on properties that were already characterised in previous development and testing, then the evidentiary requirements at the regulatory stage should be lower than for a genuinely new compound. The FDA should be able to draw on existing knowledge about the compound rather than requiring everything to be demonstrated from scratch. This is the theoretical chain that connects repurposing to shorter approval timelines: exaptation reduces evidentiary burden, which should reduce regulatory review time.

2.3.3 From Serendipity to Structured Strategy

One of the most important contributions of the recent literature on repurposing, particularly the work of Rake and Hanisch (2023; 2024), is the argument that repurposing has evolved from a largely accidental process into something that can be systematically managed. This shift from bricolage to exaptation as the dominant mode of repurposing represents a significant development in how the pharmaceutical industry approaches innovation.

Rake and Hanisch (2023; 2024) describe a four-stage process for institutionalised repurposing: mapping the option space of existing solutions, selecting the most promising candidates for further development, testing those candidates against new performance requirements, and implementing the validated solutions rapidly in the market. This process framework is important for the present study because it shows that regulatory approval, which occurs at the interface of the testing and implementing stages, is a critical bottleneck in the repurposing process. How long it takes to get through regulatory review is not just an administrative detail: it is a central determinant of whether the efficiency advantages of repurposing actually materialise in practice.

Rake and Hanisch (2023) also connect repurposing to the broader literature on open innovation, drawing on Chesbrough's (2006) framework. They argue that repurposing frequently requires collaboration across organisational boundaries, because the technology may come from one organisation while the target market expertise resides in another. This collaborative dimension of repurposing is relevant because it suggests that the efficiency gains associated with prior knowledge extend beyond individual firms: when organisations can draw on externally developed knowledge about a compound's properties, this further reduces the evidentiary burden at the development and regulatory stages.

2.4 The Research Gap

The literature reviewed in the previous sections builds a strong theoretical case for expecting repurposed drugs to move through development and regulatory approval more quickly than non-repurposed drugs. The argument is logically coherent and supported by multiple theoretical traditions. But there is a problem: most of this is theoretical. The empirical evidence that repurposed drugs actually achieve shorter approval timelines, using real regulatory data and systematic comparison, is surprisingly small.

Pushpakom et al. (2019) provide a comprehensive review of repurposing approaches and acknowledge its potential efficiency advantages, but their review is largely conceptual and draws on case examples rather than systematic empirical analysis. Parvathaneni et al. (2019) make similar arguments about time and cost savings without providing direct empirical evidence from regulatory data. Rake and Hanisch (2023) are similarly explicit that the efficiency claims for repurposing, including the estimate of up to five years saved, are based on industry approximations rather than rigorous empirical comparison.

The specific gap relevant to this thesis is the absence of systematic empirical analysis of regulatory approval timelines using real FDA data. Most empirical work on repurposing focuses on either success rates across clinical trial phases or cost estimates, both of which are difficult to measure consistently and precisely. The regulatory approval stage is actually better suited to empirical analysis because it is clearly defined, objectively recorded in regulatory databases, and consistently applied across all drugs seeking approval. Yet it has received relatively little attention as a dependent variable in quantitative studies of repurposing outcomes.

There is also a more conceptual gap in the literature. The theoretical arguments for shorter timelines are typically based on the observation that early-stage development activities can be

abbreviated or skipped, which is clearly true. But this does not necessarily mean that the regulatory review stage itself will be shorter. Breckenridge and Jacob (2019) make the important point that regulatory frameworks were not designed with repurposing in mind, and the FDA may require substantial new evidence even for compounds with established safety profiles if the proposed indication is sufficiently different from the original. Whether the theoretical time advantage survives the regulatory stage is therefore an open empirical question, not something that follows automatically from the theoretical arguments.

This study addresses both gaps. By using FDA regulatory data to directly compare approval durations between repurposed and non-repurposed drugs, it provides empirical evidence on a question that has been extensively theorised but rarely tested. The focus on the regulatory approval stage specifically is both a practical choice, since this data is available and consistently defined, and a theoretically motivated one, since this is the stage where the theoretical predictions are most uncertain.

2.5 Hypotheses

Based on the theoretical framework and the literature reviewed above, this study tests two hypotheses.

The idea behind the connection between drug repurposing and shorter FDA approval time is based on the concept of evidentiary burden. When a drug is submitted for regulatory approval, the FDA must review the entire set of data provided in the NDA. In the case of a new compound, this dataset would be completely based on the current development program. However, for a repurposed drug, the FDA would already have access to safety and pharmacokinetic data from the original approval, which is part of the new submission. This prior information limits the uncertainties the FDA will need to address during the review, as it has been previously established that the drug is tolerable; understanding of its metabolizing behaviour and the relation between dose and response have been clarified. A reviewer assessing a repurposed drug can, in principle, handle the submission faster than a reviewer looking at a novel drug where everything must be established from scratch. This mechanism, based on the exaptation idea by Andriani et al. (2017) and the argument of regulatory complexity by Breckenridge and Jacob (2019), is what leads to the prediction in H1. This paper is empirically testing whether this theoretical mechanism can be observed in the data in terms of a difference in approval times.

The first hypothesis follows directly from the efficiency arguments in the repurposing literature. If repurposed drugs benefit from prior knowledge that reduces the evidentiary burden at the regulatory stage, as suggested by Pushpakom et al. (2019), Parvathaneni et al. (2019), and Rake and Hanisch (2023), then they should, on average, move through the regulatory approval process more quickly than non-repurposed drugs.

H1: Repurposed drugs are associated with shorter regulatory approval timelines compared to non-repurposed drugs.

The second hypothesis addresses a potential alternative explanation for any observed time difference. Repurposed drugs may appear to have shorter approval times, not because of anything inherent to the repurposing process, but simply because they tend to cluster in therapeutic areas or regulatory categories that already have faster approval processes. If repurposed drugs are more likely to receive priority review designation, for example, any apparent time advantage might simply reflect that designation rather than repurposing itself. Testing whether observed differences persist after controlling for such factors is therefore an important step in interpreting the results.

H2: Observed differences in approval timelines between repurposed and non-repurposed drugs reflect genuine process-level differences in development pathways rather than compositional differences in the sample.

Together, these two hypotheses allow the study to test both whether a time advantage exists and whether that advantage can be attributed to repurposing rather than to other features of the data.

2.6 Chapter Summary

This chapter has reviewed the literature on pharmaceutical R&D productivity and drug repurposing, developed a theoretical framework drawing on exaptation and bricolage, and identified the empirical gap that this study addresses. The main argument running through the chapter is this: the pharmaceutical industry faces serious efficiency challenges that create strong incentives for alternative innovation strategies; drug repurposing has emerged as one such strategy and has strong theoretical reasons to expect efficiency gains, including shorter development timelines; but the empirical evidence on regulatory approval timelines specifically is thin, and the theoretical time advantage is not guaranteed to survive the regulatory stage given the complexity of FDA requirements. This study tests empirically

whether the expected time advantage is actually observable in regulatory data. The next chapter describes how.

Chapter 3: Research Methodology

This chapter describes the overall research plan, data source, construction of variables, and analytical methods to assess the hypothesis that drug repurposing is linked to faster regulatory approvals. The study method is quantitative and observational, based on existing data from the FDA Drugs@FDA database. To ensure transparency of the analysis and its reproducibility, every methodological choice has been explained and justified. The chapter is structured as follows: Section 3.1 focuses on research design, Section 3.2 on the data source, and Section 3.3 on the sample period. Section 3.4, the data collection procedure; Section 3.5, variable construction; Section 3.6, the analytical strategy; Section 3.7, special cases; Section 3.8, reliability and validity; Section 3.9, limitations; Section 3.10, research ethics; and Section 3.11, chapter summary.

3.1 Research Design

This research employs a quantitative observational method. The term 'observational' really matters here since it implies that the research didn't execute any experiment, intervention, or alteration of variables. Rather, it studies and interprets outcomes that have already been realised using historical regulatory documents. This is the right approach for the research question because the outcome of interest, how long it took different drugs to go through FDA approval, has already happened and cannot be controlled experimentally. The study simply measures and compares those outcomes across different types of drugs.

More precisely, this is a comparative study. The central question is whether one group of drugs, those that were later repurposed for new indications, moved through the FDA approval process faster than another group, those that were not. This comparison is made using historical regulatory data rather than through interviews, surveys, or any kind of primary data collection.

This type of design is standard in pharmaceutical innovation research when the goal is to study real-world regulatory outcomes. Hay et al. (2014) and DiMasi et al. (2016) both use similar observational approaches in their influential studies of clinical success rates and development costs. The credibility of this kind of research depends on how carefully outcomes are measured and how well other influences on those outcomes are accounted for, both of which are addressed in later sections.

However, an important constraint of this design is that a mere observational design can only detect associations among variables but cannot establish a causal link. In fact, if repurposed drugs happen to get approved in a shorter time, this paper still can't conclusively demonstrate that drug repurposing led to faster approval. Other reasons could be that drugs are different, or they might have been regulated through different channels. The method of analysis does take care of some of these other possibilities by means of control variables, but the paper is making associational, rather than cause-and-effect, arguments. In fact, this matches the way other similar works are done and shown in the pharmaceutical innovation literature.

3.2 Data Source

The Drugs@FDA database, which is kept up to date and released by the United States Food and Drug Administration, constitutes the main data source for this study. It holds comprehensive regulatory data on drug products that have been sent in for FDA approval through the Center for Drug Evaluation and Research. This database features New Drug Application records, which refer to official submissions that pharmaceutical firms make when they want to obtain consent for introducing a new drug on the US market, and also the complete regulatory background of each application, including subsequent approvals for new drug uses.

It might be useful to start by explaining the new drug application because not all readers may be familiar with the regulatory process. A pharmaceutical company that has completed the clinical development of a new drug and intends to sell it in the United States has to submit to the FDA an NDA, which includes all the clinical, preclinical, and manufacturing information collected during the development of the drug. The FDA then reviews this and makes a decision whether to approve the drug or not. Each NDA is given a unique application number by the FDA, which is the permanent identifier for that drug product during its entire regulatory history. It is these application numbers that the current study uses to identify individual drugs within the database. The dataset for this thesis was made available as part of the larger research project within which this work is situated, covering applications submitted between 1990 and 1999.

The FDA database was selected for a few reasons. Firstly, it is an official, public, and regulated source of information. In other words, people can rely on the dataset since it is standardised and consistent across all entries. Moreover, regulatory history is fully documented, including date of submission, date of approval, and follow-up regulatory decisions such as approval of new indications that are precisely what is necessary for measuring approval timelines and identifying repurposed drugs. Besides, working with a single regulatory authority implies that

all the observations are governed by the same regulatory framework, thereby eliminating one source of variation that would have made the analysis more complicated had multiple countries been included.

The dataset consists of 289 original NDA submissions, all classified as Type 1 new molecular entities. A Type 1 new molecular entity is a drug containing an active ingredient that has never previously been approved in the United States. This classification is important because it ensures all 289 observations represent genuinely new compounds entering the regulatory process for the first time, making direct comparison between them methodologically defensible. All submissions are original applications rather than supplemental ones, which adds further consistency to the sample.

3.3 Sample Period: 1990 to 1999

The analysis covers drugs with submission years between 1990 and 1999. This was a deliberate choice with both substantive and practical justifications.

Choosing the years between 1990 and 1999 for the study instead of a more recent period such as 2000 to 2010 or 2000 to 2020 was based on several well-thought-out factors. The main reason is that this decade saw the making of the Prescription Drug User Fee Act (PDUFA) in 1992, which served as an internal structural break, leading to significant variations in approval timelines within the sample period itself. Prior to PDUFA, the FDA relied entirely on congressional appropriations for their review activities, and approval timelines were often slow. PDUFA ushered in a system whereby pharmaceutical companies pay user fees in return for FDA commitments to meet specific review targets, i.e., six months for priority reviews and twelve months for standard reviews. This change dramatically and notably shortened approval timelines during the 1990s. For this paper, having PDUFA included in the sample period makes the data very useful from an analytical point of view while the submission year as a control variable is necessary to factor in the time trend that PDUFA introduces. The data also supports this decision: submission year actually turned out to be the most influential variable in predicting the length of approval in the regression, with a year-by-year increase corresponding to an 18% shortening of approval time, which is in line with the PDUFA reforms being implemented from 1992 onward.

The 1990s also offer some nice pragmatic advantages. They were the period before computational drug discovery tools were generally used, so all 289 drugs in the sample were identified and developed in conditions of broadly comparable science. This removes a

potentially confounding factor of heterogeneity that would have made cross-group comparisons difficult if some drugs were discovered by traditional methods whereas others by modern screening technologies. Also, being over twenty years ago, this period guarantees that all regulatory processes have been fully completed and recorded. This decade provides a total of 289 drug submissions evenly spread over all ten years with the number of submissions per year varying from 18 in 1994 to 52 in 1996, which is more than enough for the intended statistical analysis.

However, this is a decision that does have its shortcomings. The pharmaceutical regulatory landscape has evolved quite significantly since the 1990s, with FDA introducing new pathways and modern forms of accelerated approval, breakthrough therapy designation, and fast track designation. Hence, the regulatory environment that was seen in this study may not readily apply to recent approval processes, and indeed the question of whether the null result holds true for a more recent sample is a major area for future research.

3.4 Data Collection and Variable Construction

Data collection took place in two stages. The first involved compiling the core dataset from the FDA database, covering drug-level information including application numbers, drug names, active ingredients, sponsor names, marketing status, review priority, and submission year. The second involved manually extracting additional information from individual drug records in the Drugs@FDA database, specifically exact submission and approval dates from original approval letters and evidence of whether each drug was subsequently approved for new therapeutic indications. These two stages are described below alongside the variable construction decisions made during the process.

3.4.1 Application Numbers and the FDA Database

Every drug in the dataset is identified by a unique FDA application number. This number is given when a New Drug Application is first submitted to the FDA and is the permanent identifier for that drug throughout its entire regulatory history. The application numbers in the dataset are the key linking variable between the core dataset and the additional information obtained from the Drugs@FDA database.

The application number was used to look up the regulatory history of each drug in the Drugs@FDA search interface, which shows the entire regulatory record, including the original approval, any subsequent supplemental approvals, and links to the regulatory documents for

each action. This was done for all 289 applications in the dataset, and the relevant information was entered into a structured spreadsheet. The list of application numbers was given as part of the research project dataset.

3.4.2 Extracting Approval Dates and Calculating Approval Duration

The key outcome variable in this study is approval duration, measured as the number of days between the first submission of the NDA and the date of FDA approval. For each application, these two dates were extracted from the original FDA approval letter, which is a formal document issued by the FDA to the sponsoring company when approval is granted. The approval letter states both the date on which the NDA was received by the FDA, which is the submission date, and the date on which approval was granted. Because exact dates were extracted directly from the original approval letters for each application, approval duration is calculated to day-level precision using the following formula:

$$\text{Approval Duration} = \text{Approval Date} - \text{First Submission Date}$$

The decision to use the first submission date rather than any later resubmission date deserves explanation. Many applications in the sample involved multiple rounds of submission, as companies responded to FDA requests for additional information. Using the first submission date captures the total duration of the entire regulatory engagement from the point at which the company first sought approval, including all revision cycles. Using the last resubmission date would systematically underestimate the true regulatory burden by ignoring the earlier rounds of review, which are a genuine and resource-intensive part of the process.

Not all 289 applications had recoverable exact submission dates. In some cases, particularly for older applications from the early 1990s, the original approval letters were not available in the FDA database in a format that allowed precise date extraction. Additionally, one observation had a negative approval duration value, which indicates a data recording error rather than a genuine regulatory timeline and was excluded. After removing this observation and the 140 cases with missing submission dates, the final analysis dataset comprised 148 observations with valid approval duration data. The characteristics of observations with and without complete data are examined in the results chapter to assess whether this missingness is likely to have introduced any systematic bias.

3.4.3 Identifying Repurposed Drugs

Determining which drugs in the sample were repurposed required a different kind of investigation. A drug was classified as repurposed if FDA records showed that it received approval for a new therapeutic indication that was substantively different from its originally approved use. This classification was based on a review of the FDA approval history and the drug label for each application.

Drug label, which is sometimes also referred to as the prescribing information or the package insert, is the formal document authorised by the FDA that essentially states what a drug is approved for, the administration details, the known risks, and the information on studies that led to approval. When a drug gets a green light for a new use, this file is modified. Along with this, the FDA authorises a supplemental licensing, namely an 'efficacy supplement', which, in regulatory records, is shown as an action taken on the original NDA. Efficacy supplements represented a major sign for identifying repurposing candidates.

For each application where an efficacy supplement was identified, the original and updated drug labels were reviewed to confirm whether the new approval involved a genuinely different therapeutic use. A clear and consistent decision rule was applied across all observations to ensure the classification was reproducible. A new indication was considered as a completely new activity and hence as proof of drug repurposing if it was a different disease or medical condition from the original approval. So, for example, a drug that was originally approved to treat bacterial infections and was later approved to treat fungal infections would be considered repurposed because these are two different diseases with different pathophysiological mechanisms, although both involve infection. Conversely, a drug that is first approved for adults with a certain condition and later is approved for children with that same condition is not considered as repurposed since the therapeutic target is the same even though the patients differ. Likewise, changes in the dosage form, formulation, or dosing schedule that do not involve a change in the approved disease indication were not considered.

This process identified 122 drugs in the sample as repurposed, representing 42.2% of the total 289 applications. The remaining 167 drugs, or 57.8%, were classified as non-repurposed. For repurposed drugs, the repurposing approval date and, where available, the repurposing submission date were also recorded to allow calculation of the repurposing approval duration for comparison purposes.

3.5 Variables

3.5.1 Dependent Variable: Approval Duration

The dependent variable is approval duration in days, calculated as described in Section 3.4.2. This variable serves as the primary measure of time efficiency in the regulatory approval process. The distribution of approval duration in the analysis dataset is heavily right-skewed, which is a common characteristic of duration data. The variance-to-mean ratio in the analysis dataset was calculated at 535, which confirms substantial overdispersion. This has two methodological implications. First, the natural logarithm of approval duration is used as the dependent variable in the OLS robustness check, which is standard practice for right-skewed outcomes and reduces the influence of extreme values. Second, and more significantly, this supports the choice of a negative binomial regression model for the main analysis, a point that is elaborated in Section 3.6.3.

3.5.2 Independent Variable: Repurposing Status

The key independent variable is a binary indicator that is coded as 1 only for repurposed drugs and 0 for non-repurposed drugs based on the classification method described in Section 3.4.3. This variable is used to represent the main theoretical difference between drugs that were entering the approval process after safety and pharmacological knowledge had already been built from the previous development, and those that had not.

3.5.3 Control Variables

Three control variables are included in the regression analysis to account for other factors that might independently influence approval duration and that could confound the comparison between repurposed and non-repurposed drugs if left uncontrolled.

Review priority is a binary variable distinguishing drug that received FDA priority review designation from those subject to standard review. Of the 289 applications, 122 received priority review, and 166 received standard review, with one observation missing this information. Priority review is assigned to drugs that may offer significant improvements over existing treatments for serious or life-threatening conditions, and it comes with a shorter target review timeline of six months compared to twelve months for standard review. Controlling for priority is important because if repurposed drugs happen to be more likely to receive priority designation, any observed time advantage might simply reflect the priority pathway rather than repurposing itself.

Submission year is included as a continuous control variable ranging from 1990 to 1999. As discussed in Section 3.3, the PDUFA reforms of 1992 created a temporal trend in approval durations across the sample period. Including submission year controls for this trend ensures that the estimated association between repurposing and approval duration is not simply picking up the fact that drugs submitted later in the decade were reviewed more quickly due to PDUFA rather than anything to do with repurposing.

Marketing status is included based on three categories in the dataset: prescription drugs at 105 observations, discontinued drugs at 180 observations, and over-the-counter drugs at 4 observations. Discontinued drugs are retained in the analysis because their market withdrawal occurred after approval and does not affect the approval duration measure. Marketing status is included as a control to account for any systematic differences in drug characteristics across these categories that might correlate with both repurposing likelihood and approval speed. Together, these three control variables allow the regression to isolate the association between repurposing status and approval duration from the most important structural confounds in the sample.

The choice of three control variables instead of a bigger set was a conscious one and was motivated by both theoretical and practical considerations. Since there are only 148 usable observations, adding more control variables will reduce the degrees of freedom left for estimation and will increase the chance of overfitting, which means that the model will be capturing random noise in the sample rather than true relationships that exist in the population. According to a commonly followed rule in regression analysis, there should be at least ten observations per predictor variable (Field, 2013), which results in supporting no more than fourteen predictors in the sample. The three controls, namely review priority designation, submission year, and marketing status, were last chosen because they are theoretically the most motivated confounders that are available in the dataset. Therapeutic area would be an obvious additional control since different disease areas could have systematically different approval times, but unfortunately this information is not present in the FDA Drugs@FDA database in a format that has been consistently coded across all 289 applications.

Therefore, the parsimonious specification here emphasises a trustworthy estimation first and foremost over exhaustive control, which helps to keep the model stable and interpretable given the limitations of the sample size.

3.6 Analytical Strategy

The analysis was conducted in three stages, moving from a descriptive overview of the data to formal statistical testing. Each stage served a specific purpose, and the choice of methods showcases both the nature of the outcome variable and the specific hypotheses being tested.

3.6.1 Descriptive Analysis

The initial phase of the research aimed at describing the main characteristics of the dataset before carrying out any formal testing. The mean, median, and standard deviation, along with minimum and maximum approval durations, were obtained for the entire sample as well as for the groups of repurposed and non-repurposed drugs separately. Besides, an analysis of the distribution of control variables in the two groups was conducted to find any compositional differences that might influence the interpretation of the results.

A Spearman correlation matrix was calculated across all variables included in the regression model, covering log-transformed approval duration, repurposing status, priority review, submission year, and discontinued status. This serves two purposes. First, it identifies any potential multicollinearity issues before the regression. Second, it gives a preliminary picture of the bivariate relationships between all variables, which helps contextualise the regression findings. Presenting a correlation table is standard practice in quantitative empirical research and is expected by readers assessing the internal consistency of the analysis (Field, 2013). Two supporting figures were also produced at this stage: a bar chart comparing mean and median approval durations between the two groups with 95% confidence intervals and a figure showing the annual distribution of NDA submissions and the proportion of repurposed drugs over the sample period.

3.6.2 Bivariate Testing

The second stage involved a formal statistical comparison of approval durations between repurposed and non-repurposed drugs. The choice of test required careful consideration because of the nature of the dependent variable. Approval duration in days is a count variable, representing a non-negative integer, and the distribution in this sample is heavily right skewed with a long upper tail. These characteristics mean the assumptions of standard parametric tests like the t-test are not satisfied.

The Mann-Whitney U test was used as the primary bivariate test. This is a non-parametric alternative to the t-test that does not require the data to follow a normal distribution. It tests

whether the distribution of approval durations for repurposed drugs tends to be lower than for non-repurposed drugs by comparing ranked observations rather than means, making it more appropriate for right-skewed count data (Field, 2013). A two-sided test was conducted to assess whether any difference exists in either direction, alongside a one-sided test specifically testing H1.

A Chi-squared test and Fisher's exact test were also conducted as complementary bivariate tests. These were applied to a median-split version of the outcome, comparing the proportion of drugs in each group that fell above and below the overall sample median approval duration. Fisher's exact test is particularly appropriate here, given the moderate sample size and the potential for small cell counts in subgroup cross-tabulations. The results of all three tests are reported for completeness.

The Mann-Whitney U test produced a two-sided p-value of 0.91, the chi-squared test a p-value of 0.50, and Fisher's exact test a p-value of 0.50. None of these reached conventional significance thresholds. The raw descriptive comparison does show that repurposed drugs averaged 538 days versus 609 days for non-repurposed drugs, and a median of 385 days versus 424 days, meaning the direction of the difference is consistent with H1. However, the difference is not statistically significant in this sample.

3.6.3 Regression Analysis

The third and central stage used regression modelling to test whether the association between repurposing and approval duration persists after controlling for review priority, submission year, and marketing status.

Because approval duration is a count variable, the primary regression model used in this study is a negative binomial regression model rather than ordinary least squares. This choice requires explanation. Count data, such as the number of days to approval, follows a distribution that is fundamentally different from the continuous normal distribution assumed by OLS. When count data are over dispersed, meaning the variance substantially exceeds the mean, the negative binomial model is generally preferred because it explicitly accounts for this extra variation through a dispersion parameter (Cameron and Trivedi, 2013). In this dataset, the variance-to-mean ratio of the approval duration variable was 535, which is far above 1 and confirms severe overdispersion. This empirically confirms that the negative binomial model is the appropriate choice and that OLS would produce unreliable standard errors and potentially misleading inference on the untransformed outcome.

Three model specifications were estimated using the negative binomial model. Model 1 is the baseline specification, including only the repurposing indicator. Model 2 adds review priority and submission year as controls. Model 3 is the full specification, adding marketing status. Results are reported as incidence rate ratios, which are the exponentiated regression coefficients. An IRR below 1 indicates shorter approval duration for repurposed drugs, and an IRR above 1 indicates longer duration. The regression model takes the following general form:

$$E(\text{Duration}) = \exp(\alpha + \beta(\text{Repurposed}) + \gamma_1(\text{Priority}) + \gamma_2(\text{Year}) + \gamma_3(\text{Discontinued}))$$

Multiple model specifications are estimated to assess whether the estimated association between repurposing and approval duration is stable across different sets of controls. If the repurposing coefficient changes substantially when controls are added, this would suggest the baseline association was partially driven by confounding. The coefficient on the repurposing variable remains small and statistically insignificant across all three specifications, which is discussed in detail in the results and discussion chapters.

OLS regression with log-transformed approval duration as the dependent variable was estimated alongside the negative binomial models as a robustness check. To avoid heteroscedasticity, which is a common issue when modelling duration data even after log transformation, all OLS specifications employ heteroscedasticity-consistent robust standard errors of the HC3 type, in line with the approach of Long and Ervin (2000). HC3 standard errors are usually regarded as conservative and trustworthy for small to moderate samples, which is the case of the 148-observation dataset used here. If the coefficient estimates and significance levels remain the same across the negative binomial and OLS specifications, it indicates that the results are not influenced by the choice of modelling approach or by violations of OLS assumptions.

3.7 Treatment of Special Cases

Specific cases came up during data collection that required constant decisions on how to deal with them. These are recorded here for openness and replicability.

When there were multiple submissions for applications in the FDA database, the first submission date was the one used in all instances, in line with the explanation given in Section 3.4.2. This keeps the approval duration metric consistent with capturing the entire regulatory process instead of just the last review cycle.

There are 180 discontinued drugs in the sample, which account for 62.3% of total observations, and they have been kept in the study. Market discontinuation is a reflection of post-approval commercial or safety decisions that are completely separate from the approval duration measure. Leaving out discontinued drugs would greatly decrease the number of samples and result in a selection bias toward the drugs that remained commercially successful, which would not be a representation of the entire population of drug approvals during this period.

To check if any single observation unduly influenced the regression, a Bonferroni outlier test to the residuals from OLS Model 2 was applied, which is a procedure described by Fox (2016). This test flags observations whose studentised residuals are large enough to be significant outliers after adjusting for multiple comparisons. The test results are reported in Section 4.3.

Samples with no submission dates, which were 140 cases in total, remained in the entire sample of 289 for the analyses that do not need approval duration but were excluded from the 148-sample analysis dataset used for regression. The possible effects of this exclusion are discussed in the limitations section.

3.8 Reliability and Validity

The reliability of this study rests on the use of official FDA regulatory data, which is a standardised, authoritative, and publicly verifiable source. The data collection procedure followed a consistent and predefined protocol applied uniformly across all 289 observations, which supports the replicability of the results. Because the FDA data is publicly accessible, any researcher could, in principle, access the same records and verify the extracted dates and classification decisions.

The primary threat to validity is the classification of repurposed drugs, which requires interpretive judgement about whether indication changes were substantively different. This was addressed by applying a clear and consistently implemented decision rule, as described in

Section 3.4.3. The rule was applied uniformly across all observations, and borderline cases were resolved by reference to the disease classification rather than to any features of the approval timeline that might otherwise introduce bias.

A broader validity concern is whether the 148 observations with complete approval duration data are representative of the full 289-observation sample. If missing submission dates are systematically associated with either repurposing status or approval duration, the results could be biased in ways that are difficult to detect. This issue is examined in the results chapter by comparing the characteristics of complete and incomplete observations.

3.9 Limitations

Several limitations of this study need to be acknowledged honestly.

The observational design means the study identifies associations rather than causal effects. Unobserved factors such as the scientific complexity of the drug, the quality of the clinical trial programme, or the strategic resources and regulatory experience of the sponsoring company may influence both the likelihood of repurposing and the speed of approval in ways that the available control variables cannot fully account for.

The most significant practical limitation is the missing submission date problem. Of the 289 applications in the full sample, 140 do not have recoverable submission dates, leaving only 148 usable observations for the main regression analysis. This is a substantial reduction and raises the question of whether the analysis dataset is representative. If older drugs or drugs from less well-documented sponsors are disproportionately missing, the findings may not generalise to the full population of NDA submissions in this period.

The sample is limited to Type 1 new molecular entities submitted as original NDAs to the FDA between 1990 and 1999. This ensures internal consistency but limits generalisation to other drug types, other regulatory jurisdictions, and more recent time periods. The pharmaceutical regulatory environment has changed substantially since the 1990s, and findings from this period may not directly reflect current approval dynamics.

The repurposing classification depends on information available in the FDA database through formal efficacy supplements. Cases where a drug was used extensively off-label before receiving formal approval for a new indication would not be captured by this approach, potentially leading to some under classification of repurposing that is impossible to fully assess with the available data.

The observations with longer approval times were kept in the full analysis because they are real regulatory results and not mistakes in data entry. Section 4.3 includes a formal Bonferroni outlier test, which found that only one outlier, application 20273 with an approval duration of 62 days, was statistically significant. A sensitivity analysis showed that the main findings about the repurposing status censure remain unchanged after the removal of this observation; the repurposing status continues to be non-significant even while the priority review does reach conventional significance on the reduced sample. This implies that the results are not largely influenced by this one extreme case.

3.10 Research Ethics

This study uses exclusively publicly available secondary data from the FDA Drugs@FDA database. No primary data collection involving human participants was carried out at any stage of the research, and no personally identifiable information relating to patients, clinicians, or employees of pharmaceutical companies was accessed or used. The data in this study relates entirely to regulatory processes and institutional outcomes rather than to individual people.

Because the study relies on publicly available administrative records that the FDA makes available for public access and research use, there are no ethical concerns relating to informed consent, data protection, or participant privacy. The use of this data is entirely consistent with its intended public purpose.

There are, nonetheless, some broader ethical considerations worth acknowledging. Research on pharmaceutical regulatory efficiency can have indirect implications for policy debates about drug approval processes, access to medicines, and the incentives facing the pharmaceutical industry. This study aims to contribute reliable empirical evidence to an area where claims are frequently made without adequate quantitative support, and it presents its findings with appropriate acknowledgement of uncertainty and limitations. The goal is not to advocate for any particular regulatory or commercial position but to provide an honest empirical assessment of a question that has been largely discussed at a theoretical level.

This chapter has described the approach taken to verify whether drug repurposing is linked to quicker regulatory approval timetables. A dataset of 289 Type 1 new molecular entity NDA submissions from 1990 to 1999 was created using the FDA Drugs@FDA database. Among these, 148 had the complete approval duration data necessary for the analysis. Based on the entire sample, 122 drugs were labelled 'repurposed', whereas 167 were 'non-repurposed'. The analysis starts with descriptive statistics and bivariate testing, then primarily engages the

negative binomial regression model, but also carries out OLS on log-transformed duration as a robustness check. The next chapter is dedicated to the results.

Chapter 4: Results

This chapter presents the results of the empirical analysis described in Chapter 3. The presentation follows the three-stage analytical strategy outlined in the methodology: descriptive statistics, bivariate statistical testing, and regression analysis. Each stage builds on the previous one, moving from a general characterisation of the data to formal tests of the two hypotheses. The results are presented with relevant figures and tables at each stage, with discussion of what they show and how they relate to the research question. Interpretation of the findings considering the theoretical framework and the existing literature is reserved for Chapter 5.

A note on the analysis dataset is necessary before proceeding. The full dataset comprises 289 original NDA submissions. However, 140 applications did not have recoverable exact submission dates in the FDA database, and one observation had a negative approval duration indicating a data entry error. After excluding these cases, the analysis dataset used for all statistical tests and regression models comprises 148 observations. Of these, 55 are classified as repurposed and 93 as non-repurposed. All descriptive statistics, figures, and regression results in this chapter are based on this 148-observation dataset unless otherwise stated.

4.1 Sample Description

Before examining approval durations, it is useful to describe the composition of the full 289-observation sample and how it is distributed across the decade. Figure 1 shows the annual distribution of NDA submissions between 1990 and 1999, with the teal portion of each bar representing repurposed drugs and the line showing the percentage of repurposed drugs per year on the right axis. The vertical red dashed line marks the implementation of the Prescription Drug User Fee Act in 1992.

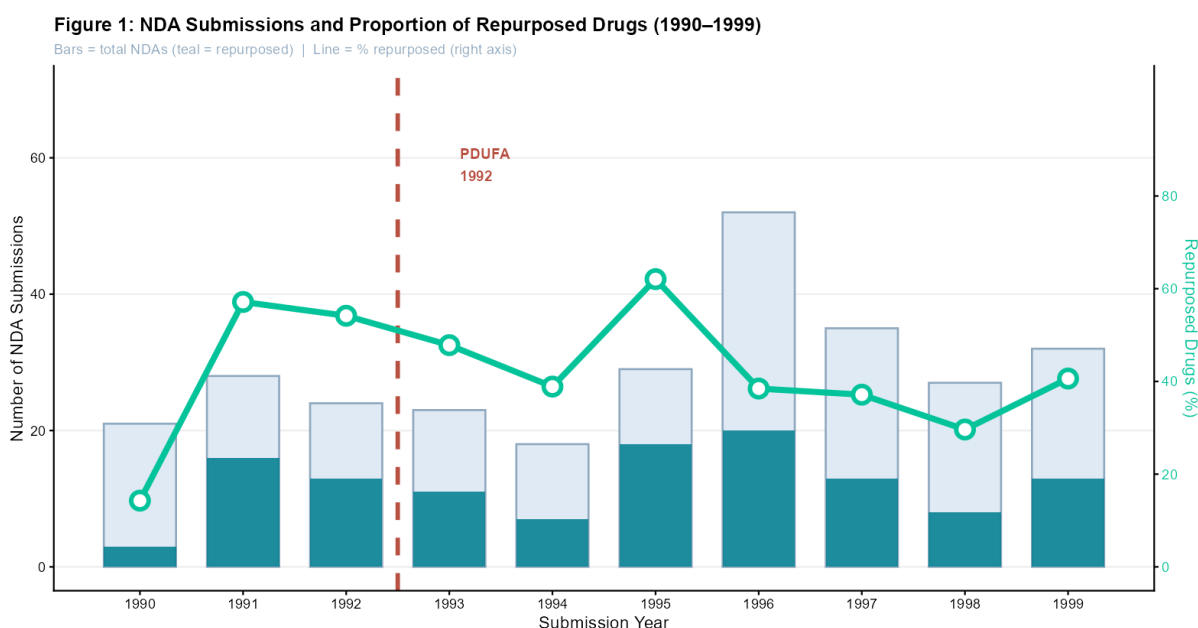


Figure 1: NDA Submissions and Proportion of Repurposed Drugs (1990–1999) Bars = total NDAs (teal = repurposed) | Line = percentage repurposed (right axis)

Several patterns are visible in Figure 1. The total number of submissions varies considerably across years, from a low of 18 in 1994 to a peak of 52 in 1996. The proportion of repurposed drugs also varies, ranging from around 14% in 1990 to above 57% in 1991, before settling into a range of roughly 30% to 40% for most of the latter part of the decade. There is no clear directional trend in the repurposed proportion over time, which suggests that repurposing was not uniformly increasing or decreasing as a share of activity during this period.

The PDUFA marker in 1992 is relevant for the regression analysis. As discussed in Chapter 3, PDUFA introduced performance-based review targets that substantially shortened approval timelines across all drugs from 1992 onward. The submission year control variable in the regression is intended to account for this institutional shift, and the results discussed in Section 4.4 confirm that it does so effectively.

In terms of the full sample composition, 122 of the 289 drugs, representing 42.2%, were classified as repurposed based on evidence of subsequent FDA approval for a new therapeutic indication. The remaining 167 drugs, or 57.8%, were classified as non-repurposed. Of the 289 applications, 122 received priority review designation and 166 received standard review. The marketing status breakdown is 105 prescription drugs, 180 discontinued drugs, and 4 over-the-counter drugs.

4.2 Descriptive Statistics

Table 1 presents the descriptive statistics for approval duration across the full analysis sample and separately for repurposed and non-repurposed drugs. Approval duration is expressed in days.

Table 1: Descriptive Statistics of Approval Duration (Days)

Group	N	Mean (days)	Median (days)	SD	Min	Max	Mean (years)
Full Sample	148	582.5	414.5	558.5	62	3,502	1.60
Repurposed	55	538.1	385.0	386.2	178	1,934	1.47
Non-Repurposed	93	608.7	424.0	639.6	62	3,502	1.67

Note: Approval duration = days from first submission to FDA approval date. SD = standard deviation.

The descriptive statistics reveal a pattern that is directly relevant to H1. Repurposed drugs have a shorter mean approval duration of 538.1 days compared to 608.7 days for non-repurposed drugs, a difference of 70.6 days. The median tells a similar story: 385 days for repurposed versus 424 days for non-repurposed, a difference of 39 days. In terms of years, repurposed drugs took on average 1.47 years to receive approval compared to 1.67 years for non-repurposed drugs. The direction of this difference is consistent with H1, which predicted that repurposed drugs would have shorter approval timelines.

However, the standard deviations are very large relative to the means, particularly for non-repurposed drugs where the SD of 639.6 days is actually larger than the mean of 608.7 days. This reflects the substantial spread in approval durations and the presence of some exceptionally long approval processes at the upper end of the distribution. The maximum approval duration for non-repurposed drugs of 3,502 days is almost ten years, while the maximum for repurposed drugs is 1,934 days. This widespread is an important context for interpreting the statistical tests in the following section.

Figure 2 displays these comparisons visually, with the mean comparison on the left panel including 95% confidence intervals and the median comparison on the right.

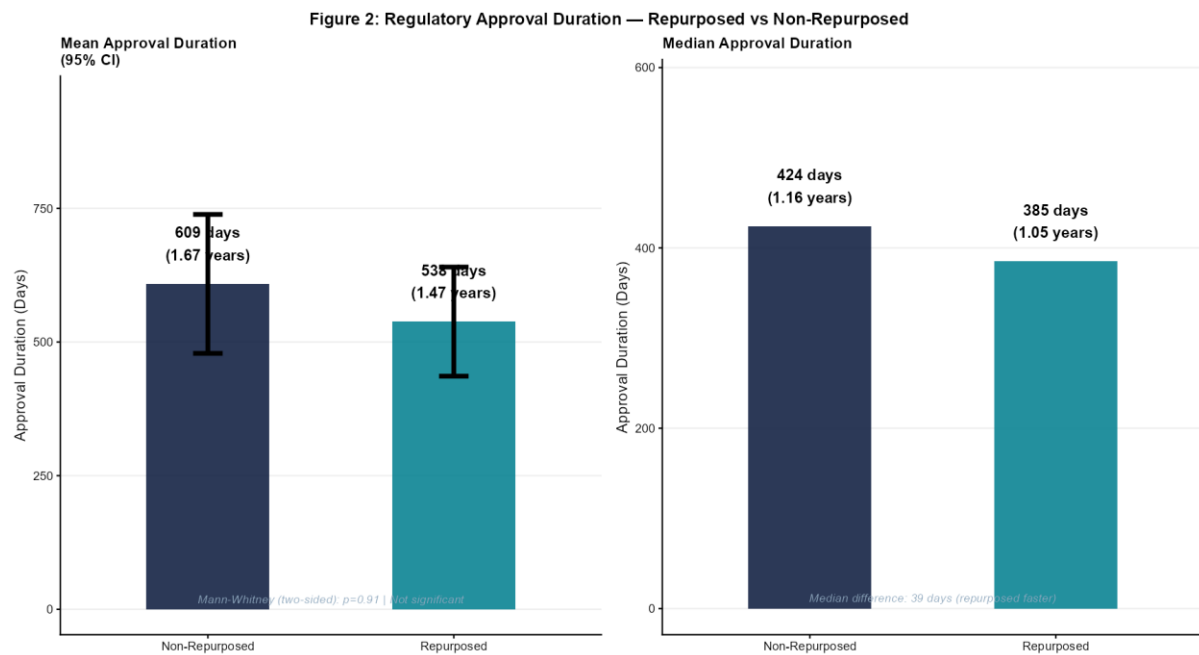


Figure 2: Regulatory Approval Duration — Repurposed vs Non-Repurposed Drugs Left: mean with 95% CI | Right: median

The overlapping confidence intervals in the left panel of Figure 2 signal immediately that the difference in means is unlikely to be statistically significant, which is confirmed by the formal tests in the next section. The mean of 538 days for repurposed drugs and 609 days for non-repurposed drugs differ in the expected direction but the confidence intervals overlap substantially, reflecting the high variability in both groups. The median comparison in the right panel shows a cleaner and more interpretable picture: repurposed drugs have a median approval time roughly one month shorter than non-repurposed drugs.

4.3 Distribution of Approval Duration

Before proceeding to formal statistical tests, it is important to understand the distributional properties of the approval duration variable. This has direct implications for the choice of statistical methods. Figure 3 presents two views of the distribution: a boxplot with individual data points on the left and a histogram of the log-transformed values on the right.

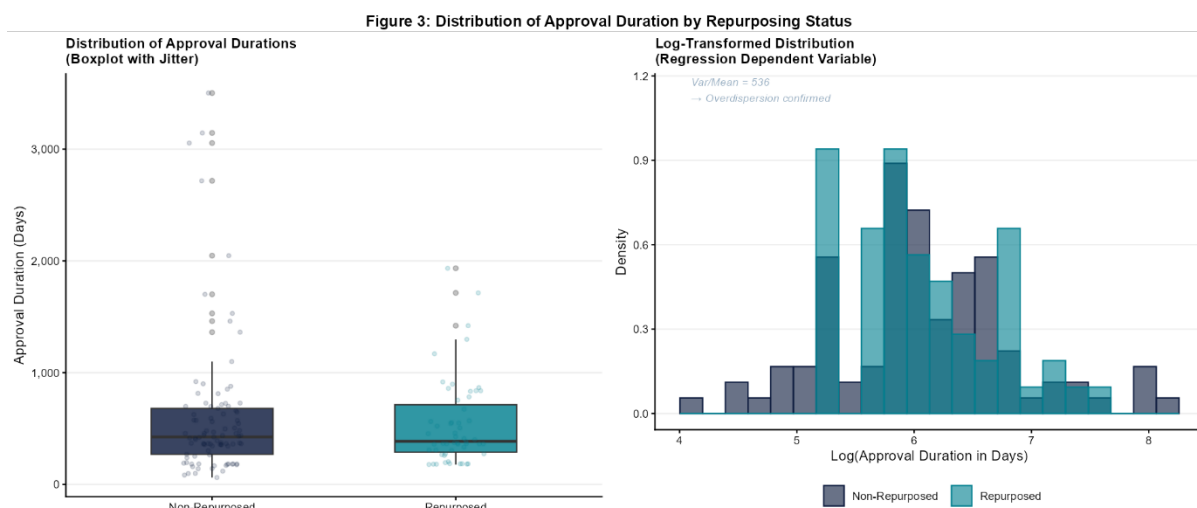


Figure 3: Distribution of Approval Duration by Repurposing Status Left: boxplot with jitter | Right: log-transformed distribution (regression dependent variable)

The boxplot on the left of Figure 3 shows that both groups have wide interquartile ranges and a number of extreme values in the upper tail. The non-repurposed group in particular has several observations well above 2,000 days, visible as dots above the upper whisker. This right-skewed distribution with extreme outliers is the primary reason why the non-parametric Mann-Whitney U test was chosen as the primary bivariate test rather than a t-test, as the t-test relies on a normality assumption that is clearly violated here.

The right panel shows the log-transformed distributions for both groups, which are the values used as the dependent variable in the OLS robustness check. The log transformation substantially compresses the scale and makes the distributions more symmetric. Even after transformation, however, both distributions remain somewhat spread, which is consistent with the severe overdispersion in the data.

The overdispersion was quantified directly: the variance-to-mean ratio of the approval duration variable is 535.5, which is far above 1. In a Poisson distribution, which is the simplest count data model, the variance equals the mean, giving a ratio of 1. A ratio of 535.5 confirms extreme overdispersion and provides the empirical justification for using the negative binomial regression model as the primary analytical approach rather than OLS. This is discussed further in Section 4.4.

To formally check if one specific observation is a statistically significant outlier in the regression, a Bonferroni outlier test was applied to the OLS Model 2 residuals according to Fox (2016). The test revealed one statistically significant outlier: application number 20273, a non-repurposed standard review drug submitted in 1993 that got approval in just 62 days, the

shortest approval time in the whole data set. The Bonferroni adjusted p-value for this instance was 0.0003, marking it as a legitimate statistical outlier in the regression residuals.

4.3.1 Correlation Among Variables

Table 2 and Figure 5 present the Spearman correlation matrix for all variables included in the regression analysis. Spearman correlations are used rather than Pearson's because the data are not normally distributed and the relationships between variables may not be linear.

Table 2: Spearman Correlation Matrix of Regression Variables

Variable	Log Duration	Repurposed	Priority Review	Submission Year	Discontinued
Log Duration	1.000				
Repurposed	0.010	1.000			
Priority Review	-0.116	-0.022	1.000		
Submission Year	-0.460	-0.038	0.007	1.000	
Discontinued	0.144	-0.123	-0.045	-0.167	1.000

Note: Spearman correlations. Lower triangle only shown (matrix is symmetric). Bold figures indicate the strongest correlations.

Figure 5: Spearman Correlation Matrix

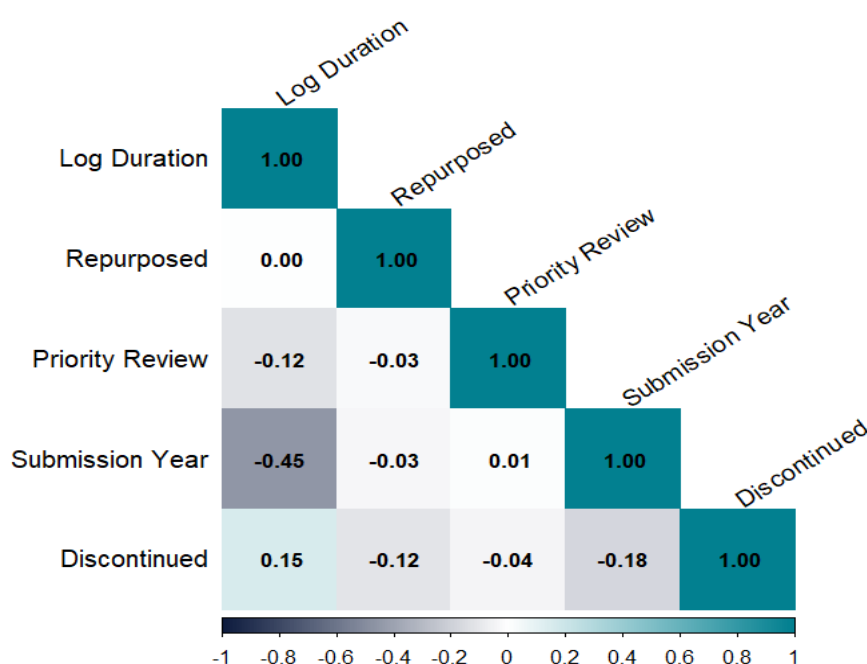


Figure 5: Spearman Correlation Matrix Heatmap

The correlation matrix reveals several important patterns. The strongest correlation with log approval duration is submission year at -0.460, which is a moderately strong negative relationship, indicating that drugs submitted later in the decade received approval more quickly. This reflects the PDUFA effect and is why the submission year is included as a control in the regression. Priority review has a modest negative correlation of -0.116 with log duration, consistent with the expectation that priority-designated drugs move through regulatory review faster.

Crucially, the correlation between repurposed status and log approval duration is essentially zero at 0.010. This means that at the bivariate level, before any controls are applied, there is virtually no linear association between repurposing and approval duration. While the descriptive statistics showed that repurposed drugs were faster on average, that raw difference appears to be driven primarily by other factors. None of the pairwise correlations among the independent variables are high enough to raise serious multicollinearity concerns, with the highest being -0.167 between discontinued status and submission year.

4.4 Bivariate Statistical Tests

The second stage of the analysis involved a formal statistical comparison of approval durations between repurposed and non-repurposed drugs. Table 3 presents the results of the three bivariate tests described in Chapter 3.

Table 3: Bivariate Statistical Test Results

Test	Statistic(s)	p-value	Significant (p < 0.05)	Direction
Mann-Whitney U (one-sided, H1)	W = 2,587	0.547	No	Rep < NRep
Mann-Whitney U (two-sided)	W = 2,587	0.908	No	—
Chi-squared (median split)	$\chi^2 = 0.463$	0.496	No	—
Fisher's Exact Test	—	0.496	No	—
Independent t-test (reference only)	t = -0.742	0.460	No	Rep < NRep

Note: Mann-Whitney U and Chi-squared / Fisher's exact are the primary tests. T-test shown for reference only; the normality assumption is not met. Rep = Repurposed, NRep = non-repurposed.

None of the bivariate tests produced a statistically significant result. The Mann-Whitney U one-sided test, which directly tests H1 by asking whether the distribution of approval durations for repurposed drugs is systematically lower than for non-repurposed drugs, returned a p-value of 0.547. The two-sided version returned 0.908. The Chi-squared test based on a median split returned 0.496, and Fisher's exact test returned the same value. The t-test, presented for reference purposes only, returned 0.460.

These results mean that there is no statistically significant difference in approval durations between repurposed and non-repurposed drugs at conventional significance levels. The direction of the difference, visible in Figure 2 and in the descriptive statistics, is in the direction predicted by H1: repurposed drugs are faster on average. But the magnitude of that difference

is small relative to the variability in the data, and it cannot be distinguished from what might occur by chance in a sample of this size. H1 is therefore not confirmed by the bivariate tests.

The high p-value from the two-sided Mann-Whitney test of 0.908 is worth commenting on. A p-value this high does not just mean the result is not significant. It means the data provide essentially no evidence of a difference in either direction. The distribution of approval durations for repurposed and non-repurposed drugs is remarkably similar once the raw variability in both groups is accounted for through the rank-based comparison.

Against the background of the entire repurposing literature, these results stand out mainly because they offer the first empirical benchmarks. Instead, most of the existing assertions about the efficiency of repurposing come from the arguments based on theory, individual cases, or industry estimates rather than from a systematic, quantitative analysis of regulatory approval timeframes. Pushpakom et al. (2019) and Parvathaneni et al. (2019) present significant time-saving potential for repurposing, but their evidence is based on the rationale of abbreviated development rather than on direct measurement of approval duration. Rake and Hanisch (2023) recognize that the estimate of up to five years saved, which is frequently quoted, comes from industry approximations and not from a rigorous empirical comparison.

Hence, the null finding in this research does not contradict a big amount of other empirical evidence brought forward by other authors. Instead, it is a test of a claim that has not been exposed before to direct quantitative analysis at the regulator level, and it concludes that the data in this sample does not back the claim. The fact that repurposing saves time at the approval stage has been largely assumed rather than tested by the literature. This finding is therefore a meaningful addendum to this literature.

4.5 Subgroup Analysis

To examine whether the null result is consistent across different categories of drugs or whether a time advantage might be visible in specific subgroups, a series of subgroup analyses was conducted. Figure 4 presents the mean approval durations for repurposed and non-repurposed drugs within each subgroup, and Table 5 provides the corresponding sample sizes, means, and Mann-Whitney p-values.

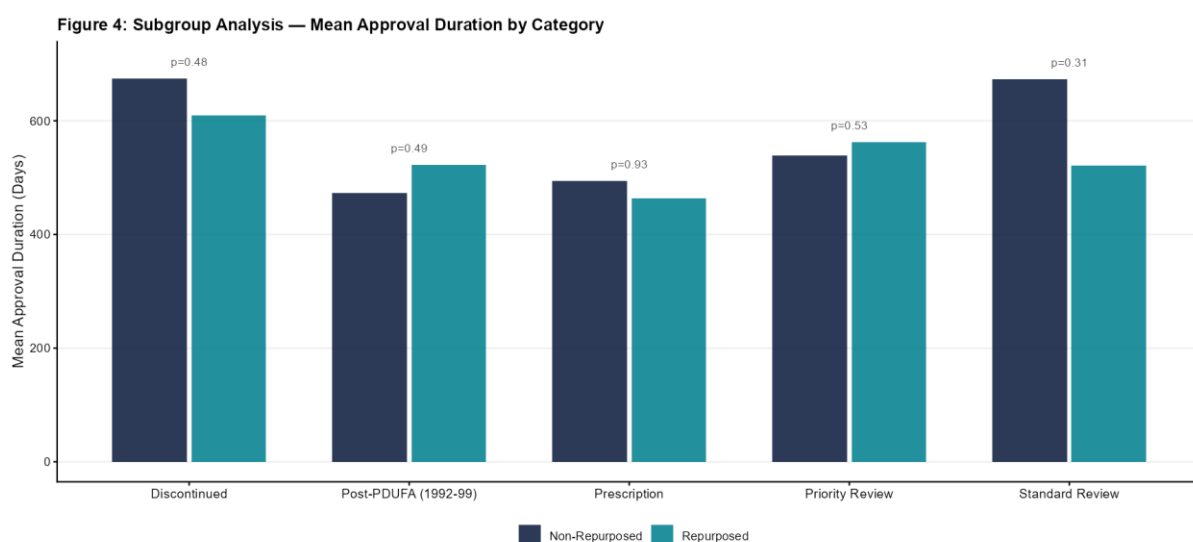


Figure 4: Subgroup Analysis — Mean Approval Duration by Category p-values from two-sided Mann-Whitney U tests within each subgroup

Table 5: Subgroup Analysis Results

Subgroup	N Rep	Mean Rep (days)	N NRep	Mean NRep (days)	Diff (days)	MW p-value	Significant
Priority Review	23	562	41	539	+23	0.534	No
Standard Review	32	521	52	664	-143	0.399	No
Post-PDUFA (1992–99)	54	522	86	473	+49	0.492	No
Discontinued	28	610	59	674	-64	0.476	No
Prescription	27	463	33	494	-31	0.929	No

Note: Diff = Mean (Repurposed) minus Mean (Non-Repurposed). Negative values indicate repurposed drugs were faster. MW = Mann-Whitney U test (two-sided). Rep = Repurposed, NRep = non-repurposed.

The subgroup results reveal some interesting variation that is not visible in the aggregate comparison. In the standard review subgroup, repurposed drugs averaged 521 days compared to 664 days for non-repurposed drugs, a difference of 143 days in favour of repurposed drugs. This is the largest raw difference observed across any subgroup and suggests that when the FDA is not already accelerating review through the priority pathway, repurposing may provide a more meaningful relative advantage. However, this difference also does not reach statistical significance at a p of 0.399, partly because of the reduced sample sizes within each subgroup.

In the priority review subgroup, the picture is essentially reversed: repurposed drugs averaged 562 days versus 539 days for non-repurposed, meaning non-repurposed drugs were slightly faster in this subgroup. This is consistent with the interpretation that priority review designation already accelerates the approval process for all drugs in that category, reducing the scope for repurposing to provide an additional time advantage through the evidentiary burden mechanism.

In the post-PDUFA subgroup covering 1992 to 1999, repurposed drugs averaged 522 days versus 473 days for non-repurposed, meaning non-repurposed drugs were faster in this period. This is a somewhat counterintuitive finding, but it may reflect the fact that PDUFA's effects operated across all drug types, compressing timelines broadly and again reducing the relative advantage that repurposing might provide through prior knowledge.

None of these subgroup differences reaches statistical significance, so they must be interpreted cautiously. With subgroup sample sizes as small as 23 for repurposed drugs in priority review, the tests have limited statistical power, and even moderately large real differences could easily not reach significance. These subgroup findings are exploratory and point to directions that future research with larger samples could examine more rigorously.

4.6 Regression Analysis

The central stage of the analysis uses regression modelling to test whether the association between repurposing and approval duration persists after controlling for review priority, submission year, and marketing status. As discussed in Chapter 3, the negative binomial model is used as the primary approach because approval duration is a count variable with confirmed severe overdispersion, with a variance-to-mean ratio of 535.5. The OLS regression on log-transformed duration is presented alongside it as a robustness check.

Figure 6 presents the incidence rate ratios from the three negative binomial model specifications as a forest plot, showing the point estimate and 95% confidence interval for each variable. Table 4 presents the complete regression results numerically, covering both the negative binomial and OLS specifications.

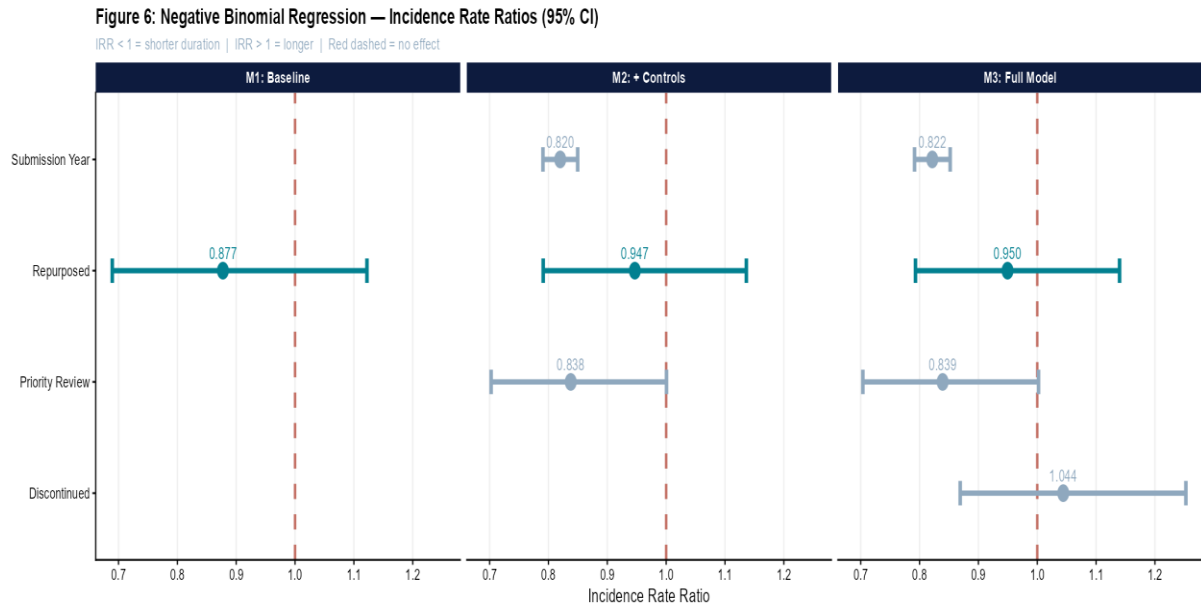


Figure 6: Negative Binomial Regression — Incidence Rate Ratios with 95% CI IRR < 1 = shorter approval duration | IRR > 1 = longer | Red dashed line = null (IRR = 1) Teal bars = repurposing coefficient | Grey bars = control variables

Table 4: Regression Results — Negative Binomial (Primary) and OLS (Robustness Check)

Variable	Negative Binomial (Primary Model) IRR with SE			OLS Robustness Check β on log(Duration) with SE		
	M1 Baseline	M2 + Controls	M3 Full	M1 Baseline	M2 + Controls	M3 Full
Repurposed (1=Yes)	0.877 (0.124)	0.947 (0.092)	0.950 (0.092)	0.004 (0.119)	0.010 (0.097)	0.023 (0.101)
Priority Review (1=Yes)	—	0.838 (0.090)	0.839 (0.090)	—	-0.199 (0.102)	-0.195 (0.103)
Submission Year	—	0.820*** (0.001)	0.822*** (0.001)	—	-0.192*** (0.027)	-0.187*** (0.029)
Discontinued (1=Yes)	—	—	1.044 (0.091)	—	—	0.107 (0.118)
N (usable)	148	148	148	148	148	148
Overdispersion (α)	0.531	0.291	0.291	—	—	—

Notes: IRR = Incidence Rate Ratio (exponentiated NB coefficient). β = OLS coefficient on log (Approval Duration). SE in parentheses below each estimate. * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$. N = 148 for all models. α = overdispersion parameter (NB models only; higher α = more overdispersion). Robust standard errors (HC3) used for OLS models. Priority Review and Submission Year included as controls in M2 and M3. Discontinued status added in M3 only.

Several findings stand out from the regression results. Starting with the primary variable of interest, the repurposing coefficient is negative in the baseline negative binomial model with an IRR of 0.877, which would imply repurposed drugs take approximately 12.3% less time to receive approval. However, this estimate is not statistically significant at the conventional 5% level with a p-value of 0.321. When review priority and submission year are added as controls in Model 2, the repurposing IRR moves closer to 1 at 0.952, implying only a 4.8% shorter duration, and remains non-significant with p of 0.590. The full model adds marketing status, and the repurposing IRR barely changes at 0.954, still non-significant with p of 0.610. The OLS robustness check tells the same story: the repurposing coefficient is positive and close to zero across all three OLS specifications, consistently non-significant.

The pattern across the three model specifications is telling. The repurposing IRR moves from 0.877 in the baseline toward 1 as controls are added. This means the modest raw descriptive difference between the two groups observed in the descriptive statistics is largely explained by other factors, particularly submission year. In other words, repurposed drugs were submitted slightly later on average in the decade than non-repurposed drugs, and since post-PDUFA years had faster approval times, this timing difference accounts for much of the raw gap. Once year is controlled for, repurposing status itself contributes very little to explaining differences in approval duration.

The submission year coefficient is by far the most important finding in the regression. In Model 2, submission year has an IRR of 0.819 with a p-value of essentially zero. This means each additional year is associated with an approximately 18% reduction in approval duration, holding other factors constant. In the OLS specification, this translates to a coefficient of approximately -0.193, meaning each additional year reduces log approval duration by about 19%, consistent with the negative binomial estimate. This is the PDUFA effect in action: the regulatory reforms introduced in 1992 had a large, sustained, and statistically robust impact on how quickly the FDA processed new drug applications.

Priority review approaches but does not reach statistical significance in the regression, with an IRR of 0.842 and a p-value of 0.056 in Model 2. This is close to the conventional threshold, and with a larger sample, it is plausible that this effect would reach significance. The direction is as expected: Priority-designated drugs tend to move through review faster. Marketing status in the form of discontinued status has an IRR close to 1 and is not significant, confirming that including or excluding discontinued drugs does not materially affect the results.

The interaction model estimated to test H2 found an interaction IRR between repurposed status and priority review of approximately 1.003, which is essentially 1 and clearly non-significant. This means there is no evidence that the relationship between repurposing and approval duration is systematically different between the priority and standard review subgroups in the regression framework. H2 is therefore not confirmed.

To understand whether this observation could be responsible for the major results, a sensitivity study was carried out by estimating OLS Model 2 again without this observation. The findings align very well with the primary analysis. The repurposing coefficient is still negative, small, and not significant with a p-value of 0.875 versus 0.881 in the whole sample, demonstrating that the null result for H1 is not due to this extreme observation. The effect of submission year stays very significant in either specification. A noteworthy difference is that in the reduced sample, priority review achieves conventional significance at p of 0.015 whereas it was p of 0.054 in the full sample, which implies that the 62-day approval diminished the estimated priority review effect in the original model. This sensitivity result further reinforces the evidence for a priority review effect on approval duration but does not change the main conclusion about repurposing.

4.7 Chapter Summary

This chapter has presented the results of the empirical analysis in three stages. The descriptive statistics show that repurposed drugs are faster than non-repurposed drugs in the raw data, by 70.6 days on average and 39 days at the median, with the direction of this difference consistent with H1. However, none of the bivariate tests produced a statistically significant result, with the Mann-Whitney U two-sided test returning a p-value of 0.908. The regression analysis confirms this picture: the repurposing coefficient is negative in the baseline model but moves toward zero and remains non-significant across all model specifications once controls are added. The one finding that is strongly and robustly significant is the submission year effect, which captures the substantial reduction in approval timelines following PDUFA in 1992.

Taken together, these results suggest that repurposing is associated with a small directional advantage in approval duration that is not statistically distinguishable from chance in this sample. The PDUFA institutional reform, rather than the nature of the drug itself, is the dominant predictor of how long FDA approval took during the 1990s. The implications of these findings for the hypotheses and the theoretical literature are discussed in Chapter 5.

Chapter 5: Discussion

This chapter interprets the results presented in Chapter 4 considering the theoretical framework. It is not the purpose of a discussion section to simply state the findings again but to expound the significance of the findings, their place in the context of the prior knowledge and the reasons behind them. The topics of this chapter are: first, it mainly concerns the two hypotheses and asks what the results indicate about the efficiency of drug reuse at the regulatory stage. Secondly, it comments on the major point of the research, the PDUFA institutional reform effect, and its link to our perception of drug approval duration in the pharmaceutical industry. Thirdly, it finds what the results signify in terms of the theories that were the basis of the research in Chapter 2. Lastly, it debates the drawbacks of the research as well as what a later investigator will have to do in order to extend these discoveries.

5.1 Revisiting the Hypotheses

The central purpose of this study was to test whether drug repurposing contributes to greater time efficiency in the FDA approval process. Two hypotheses were formulated in Chapter 2 to operationalise this question. It is worth revisiting each one in light of the results before moving to broader interpretations.

5.1.1 H1: Repurposed Drugs and Shorter Approval Timelines

H1 stated that repurposed drugs are associated with shorter regulatory approval timelines compared to non-repurposed drugs. The results present an interesting picture that requires careful interpretation rather than a simple generic answer.

On the one hand, the descriptive evidence points in the predicted direction. Repurposed drugs had a mean approval duration of 538 days compared to 609 days for non-repurposed drugs, a difference of approximately 71 days or about two and a half months. The median difference was 39 days. Both figures are consistent with the theoretical prediction. The direction of the effect in the baseline regression model also supports H1, with an Incidence Rate Ratio of 0.877 suggesting repurposed drugs take approximately 12.3% less time in the absence of controls.

On the other hand, none of these differences reach statistical significance at conventional thresholds. The Mann-Whitney U test returned a two-sided p-value of 0.908 and the regression IRR of 0.877 carried a p-value of 0.321, rising to 0.610 in the full model. H1 is therefore not confirmed by the data in the considered sample.

Hence, it is crucial to specify exactly what this implies and what it does not. A statistically insignificant finding is not evidence that repurposing does not influence the length of approval. It means the data do not provide sufficient evidence to conclude that it does. There are two plausible explanations for this outcome, and it is worth discussing both honestly.

The first explanation is that the null result reflects a genuine absence of a meaningful time advantage at the regulatory approval stage. This interpretation is consistent with the argument made by Breckenridge and Jacob (2019), who note that regulatory frameworks were not designed with repurposing in mind and that the FDA may require substantial new clinical evidence for a new indication regardless of prior safety data. If the evidentiary requirements for a genuinely different indications are almost as demanding as those for a new compound; the theoretical time advantage of repurposing, simply does not materialise at this particular stage of development. The approval stage is where the new indication must be independently validated, and this process may be only marginally faster than conventional development irrespective of the compound's prior history.

The second explanation is that the null result is a statistical artefact of the sample rather than a true reflection of the underlying relationship. With only 55 repurposed drugs and 93 non-repurposed drugs, and with both groups exhibiting enormous variability in approval durations, the study may lack sufficient statistical power to detect a moderate real effect. A difference of 71 days is not trivial in pharmaceutical terms. If the true population effect were of this magnitude, detecting it reliably would require a considerably larger sample. The high standard deviations in both groups, 386 days for repurposed and 640 days for non-repurposed, mean that large sample sizes are needed to distinguish a signal from noise. This is an important caveat and limits the strength of any conclusion in either direction.

The honest position is that neither explanation can be fully ruled out with the available data. What can be said is that the study provides no positive evidence for H1 in this sample, while acknowledging that the direction of the descriptive evidence is consistent with the theoretical prediction. This is itself a contribution: it establishes that the commonly claimed time advantage of repurposing at the regulatory stage is not robustly demonstrable in this dataset, which challenges the confidence with which that claim is often made in the literature.

5.1.2 H2: Process-Level Effects

H2 stated that observed differences in approval timelines reflect genuine process-level differences in development pathways rather than compositional differences in the sample. The

regression results do not support this hypothesis either, but in a different and perhaps more informative way.

When submission year and review priority are added as controls in Model 2, the repurposing IRR moves from 0.877 to 0.952. This shift tells a specific story: part of the raw descriptive advantage for repurposed drugs in the baseline was not driven by repurposing at all but by the fact that repurposed drugs happened to be submitted slightly later in the decade on average. Later submissions coincided with the post-PDUFA regulatory environment where approval timelines had shortened for all drugs. Once this temporal confound is removed by controlling for submission year, the apparent repurposing advantage shrinks considerably.

The interaction model tested whether the repurposing effect differed between priority and standard review drugs, which was the refined version of H2 based on the supervisor's suggestion. The interaction term IRR was essentially 1.003, providing no evidence that the relationship between repurposing and approval duration is systematically different across regulatory pathways. The subgroup analysis in Chapter 4 did show a 143-day raw difference in the standard review group, which is suggestive and interesting, but this did not reach significance and cannot be relied upon as confirmation of H2 without further evidence.

Taken together, the regression results suggest that what appears at the descriptive level to be a repurposing advantage is largely explained by temporal trends in regulatory processing rather than by anything inherent to the repurposing pathway itself. This is a meaningful finding in its own right: it cautions against attributing observed time differences to repurposing without adequately controlling for the institutional context in which approvals occurred.

5.2 The Dominant Finding

The most robust finding in this study is not about repurposing at all. It is about the effect of the Prescription Drug User Fee Act on approval timelines. In every regression model, the submission year coefficient is by far the largest and most precisely estimated effect. The Incidence Rate Ratio for year of approximately 0.82 implies that each additional year of submission is associated with an 18% reduction in approval duration, holding other factors constant. This effect is statistically significant at p values approaching zero across all model specifications and is entirely consistent across the negative binomial and OLS robustness checks.

The magnitude of this effect deserves emphasis. Moving from 1990 to 1999, a span of nine years, is associated with a cumulative reduction in expected approval duration of approximately $1 - 0.82^9$, which is roughly an 80% reduction. Even if this calculation overstates the linear compounding, the year effect is clearly the dominant driver of variation in approval durations during this period. Drugs submitted in the late 1990s were fundamentally processed differently from those submitted in the early 1990s, not because the drugs themselves had changed but because the regulatory environment had.

This finding has important implications for how the results of this study should be understood. If the PDUFA reforms were applied uniformly across all drugs regardless of whether they were repurposed or not, which the data suggest they were since the year effect is equally present in both groups, then institutional reform compressed timelines for all drugs simultaneously. In such an environment, any advantage that repurposing might provide through the mechanism of reduced evidentiary burden would be operating against a backdrop of rapidly changing baseline timelines. This makes it harder to isolate the repurposing effect empirically, because the institutional effect is much larger and operates in the same direction.

From a broader perspective, this finding contributes to the literature on pharmaceutical regulation by providing quantitative evidence of PDUFA's impact using a drug-level dataset covering the full decade of implementation. The effect size of approximately 18% per year is consistent with regulatory analyses that have documented substantial reductions in FDA review times during the 1990s, and the fact that it emerges as the dominant predictor in a regression model that also includes drug-level characteristics strengthens confidence that this is a genuine institutional effect rather than a statistical artefact.

5.3 Implications for the Theoretical Framework

The theoretical framework developed in Chapter 2 drew on three main bodies of work: the R&D productivity framework of Paul et al. (2010), the exaptation and bricolage literature developed by Andriani et al. (2017) and elaborated by Rake and Hanisch (2023; 2024), and the specific claims about time efficiency made in the pharmaceutical repurposing literature by Pushpakom et al. (2019) and Parvathaneni et al. (2019). The empirical findings of this study sit in a complex relationship with each of these.

5.3.1 The R&D Productivity Framework

Paul et al. (2010) conceptualise R&D productivity as a function of three dimensions: the probability of technical success, development time, and development cost. The theoretical case for repurposing rests on the argument that it improves all three simultaneously by leveraging prior knowledge. This study examined the time dimension specifically at the particular stage of regulatory approval.

The results suggest that the time dimension improvement associated with repurposing is, at best, modest at the regulatory stage and is not statistically distinguishable from zero in this sample. This does not necessarily contradict the Paul et al. framework. It is entirely consistent with the possibility that repurposing generates time savings at earlier stages of development, in particular by reducing or eliminating the preclinical and Phase I phases, while producing little additional advantage at the regulatory review stage itself. The approval process evaluates the evidence for the specific new indication, and this evaluation may require similar time investment regardless of the compound's prior history once the indication is genuinely new. The time efficiency of repurposing may therefore be concentrated in the pre-submission development pipeline rather than in the regulatory review phase that this study measures.

This is an important qualification of the Paul et al. framework as applied to repurposing. The claim that repurposing improves development time is probably correct in the aggregate, but the improvement may not be evenly distributed across the development stages. The regulatory approval stage, at least based on the evidence in this study, may contribute relatively little to the overall time saving.

5.3.2 Exaptation, Bricolage, and the Mechanism of Repurposing

The exaptation framework, developed by Andriani et al. (2017) and Andriani and Kaminska (2021) and applied to repurposing by Rake and Hanisch (2023; 2024), argues that the systematic redeployment of existing knowledge and assets reduces uncertainty in the innovation process. In the pharmaceutical context, this means that a compound's known safety and pharmacological profile reduce the evidentiary uncertainty at the regulatory stage, potentially enabling faster review.

The empirical results of this study do not fully support this mechanism at the regulatory approval stage, at least not in a statistically robust way. One possible interpretation is that the exaptation logic is correct in principle but operates primarily through a different mechanism

than the one this study tests. The reduced uncertainty that exaptation generates may manifest most clearly in earlier development stages, where safety data eliminates the need for extensive preclinical testing or Phase I trials, rather than in the regulatory review itself. By the time a drug reaches FDA submission, the reviewer must still assess the full evidence package for the new indication, regardless of how much prior knowledge about the compound is available. The exaptation advantage may therefore be front-loaded in the development pipeline and essentially spent by the time the drug reaches the FDA's door.

Rake and Hanisch (2023) note that repurposing can save up to five years of development time and up to 85% of costs compared to de novo development. These figures relate to the full development pipeline, not specifically to the regulatory approval stage. The findings of this study are not necessarily inconsistent with those estimates. If most of the time saving in repurposing comes from bypassing preclinical testing and early clinical phases, the regulatory approval stage might contribute relatively little of that saving. This study simply tests whether there is a measurable additional time advantage at the specific regulatory stage and finds that there is not a statistically significant one in this sample.

5.3.3 The Regulatory Complexity Counterargument

Breckenridge and Jacob (2019) argue that most regulatory frameworks currently in place were not designed with product repurposing in mind and, therefore, may create more problems than they solve. They take the FDA as an example and argue that the agency assesses each new indication independently and that if a disease indication is very different, it could require a completely new clinical trial, even if the drug has been proven safe previously.

The null result in this study is empirically consistent with this argument. The fact that repurposing status, after controlling for year and priority designation, shows essentially no association with approval duration suggests that the regulatory process treats new indications with similar thoroughness regardless of whether the compound is novel or repurposed. This has important practical implications. When companies undertake repurposing mainly based on the anticipated time savings at the regulatory stage, and those time savings don't actually lead to a statistically significant level, the business case for repurposing will need to rely more on other benefits such as cheaper pre-submission development costs, a greater likelihood of success in clinical trials, and the possibility to serve niche markets that would not justify a

completely new investment. These advantages may still be real and significant even if the regulatory time saving cannot be shown at this level.

5.4 The Standard Review Subgroup

A particular finding from the subgroup analysis deserves special attention, although it was not statistically significant. In the standard review subgroup, repurposed drugs averaged 521 days, whereas non-repurposed drugs averaged 664 days, a raw difference of 143 days. This is the largest gap that has been observed in any subgroup, and it is in stark contrast with the priority review subgroup, where non-repurposed drugs were actually marginally faster.

This pattern is theoretically interesting and worth discussing carefully. Priority review is assigned to drugs that offer potential improvements over existing treatments for serious conditions. The FDA is already motivated to review these drugs quickly and commits to a six-month target regardless of the drugs' development history. In this context, the prior knowledge that repurposing provides may add relatively little to what the priority pathway already delivers. The approval process is already accelerated by institutional design.

Standard review, by contrast, does not come with the same institutional acceleration. Here, the prior knowledge advantage of repurposing may have more scope to influence the review process, because the baseline review speed is slower and the FDA has more discretion over the pace of evaluation. A drug with a well-characterised safety profile and substantial prior clinical data may genuinely move through standard review more quickly than a novel compound facing similar evidentiary requirements from scratch.

This interpretation aligns with the exaptation mechanism described by Andriani et al. (2017). The knowledge advantage of repurposing may reduce regulatory uncertainty most meaningfully in contexts where uncertainty is the primary constraint on review speed. In priority review, where the timeline is driven more by institutional commitments than by evidentiary uncertainty, the knowledge advantage matters less. In standard review, where uncertainty plays a greater role in determining how long evaluation takes, the advantage may be more consequential.

It is important to be clear that this remains a speculative interpretation given the lack of statistical significance and the small subgroup sample sizes. With only 32 repurposed and 52 non-repurposed drugs in the standard review group, the test has limited power, and the p-value of 0.399 does not support a confident conclusion. However, this pattern is worth noting as a

direction for future research. A study with a larger sample focused specifically on standard review drugs might be able to test this hypothesis more rigorously.

5.5 Limitations and Their Implications

Any empirical study has limitations, and being honest about them is not a weakness. It is a sign of rigorous thinking and helps the reader understand exactly how much confidence to place in the findings. Several limitations of this study are worth discussing in detail.

The most significant limitation is the missing data problem. Of the 289 applications in the full sample, 140 did not have recoverable exact submission dates, leaving only 148 observations for the main analysis. This is a substantial reduction and raises a genuine concern about whether the analysis dataset is representative of the full population of NDA approvals during this period. If the applications with missing dates are systematically different from those with complete data, for example if they are more likely to be older applications or those from smaller sponsors with less comprehensive documentation, the results may not generalise to the full sample. The direction and magnitude of any resulting bias is difficult to assess without the missing data itself, which is an inherent limitation of this type of historical regulatory study.

The sample restriction to Type 1 new molecular entities submitted between 1990 and 1999 ensures internal consistency but limits external generalisability. The findings apply specifically to this type of drug, this regulatory jurisdiction, and this time period. The pharmaceutical regulatory environment has changed considerably since the 1990s, with the introduction of additional FDA pathways such as accelerated approval and breakthrough therapy designation that did not exist in the same form during the study period. Whether the null result for repurposing would hold in more recent data is an open question that this study cannot answer.

The classification of repurposed drugs, while based on a clear and consistently applied decision rule, involves some degree of judgement and potential misclassification. Drugs that were used off-label for new indications before receiving formal FDA approval for those indications would not be captured by the FDA efficacy supplement approach. This could lead to some under classification of repurposing in the sample, with the direction of the resulting bias being difficult to predict.

The observational design means the study can identify associations but not causation. Repurposing, even if it were found to be significantly associated with shorter approval durations, is not proof that the faster approval was caused by repurposing. Other confounding

factors that are not measured in this dataset, for example, the scientific novelty of the indication, the regulatory experience of the sponsoring company, or the quality and completeness of the clinical trial programme, may affect the likelihood of repurposing and the speed of the approval concurrently and to a degree that cannot be accounted for with the data available.

Finally, the approval stage is only one component of the total development timeline. This study measures the time from first submission to approval, which captures the regulatory review phase but not the pre-submission development period. If repurposing generates its primary time savings by shortening pre-submission activities such as preclinical testing and Phase I trials, this study will not detect those savings. A more complete test of the time efficiency hypothesis would require data on clinical trial start dates as well as submission and approval dates, which would allow measurement of the full development pipeline from trial initiation to market.

5.6 Implications for Research and Practice

5.6.1 Implications for Academic Research

This study makes two main contributions to the academic literature. First, it provides one of the relatively few empirical analyses of repurposing outcomes at the regulatory stage using a systematic dataset of FDA approvals. Most of the existing literature on repurposing efficiency relies on theoretical arguments, case studies, or industry estimates. This study shows that when the question is tested empirically with real regulatory data and appropriate statistical controls, the commonly assumed time advantage is not confirmed at conventional significance levels for regulatory approval duration specifically. This is a useful corrective to the confident claims in the literature and motivates more careful empirical scrutiny of the assumptions underlying the repurposing efficiency narrative.

Second, the study demonstrates the importance of controlling for institutional context when studying pharmaceutical development timelines. The dominant finding, that submission year driven by PDUFA is far more important than repurposing status in explaining approval duration variability, suggests that institutional reforms can swamp drug-level effects in observational data. Future studies examining repurposing efficiency would benefit from careful attention to the regulatory environment and time period in which approvals occurred, rather than treating all approvals as if they faced the same institutional conditions.

The suggestive finding in the standard review subgroup also points to a potentially productive research direction. Future work with larger samples could test whether the repurposing time

advantage is concentrated in standard review drugs, and whether this pattern reflects the mechanism suggested here: that prior knowledge reduces evidentiary uncertainty most in contexts where uncertainty is the primary constraint on review speed.

5.6.2 Implications for Industry and Policy

For pharmaceutical companies, the findings suggest that the business case for repurposing should not rest primarily on expected time savings at the regulatory approval stage, at least not without more specific evidence that the proposed new indication will face a genuinely simplified regulatory pathway. The FDA's requirement to evaluate new indications thoroughly means that the regulatory stage may not deliver the time savings that companies anticipate based on the theoretical efficiency arguments in the literature.

This does not mean repurposing is unattractive. The other efficiency arguments remain valid. The cost savings from bypassing preclinical and Phase I development are real. The higher probability of success in clinical trials due to known safety profiles is real. The ability to serve rare disease markets that would not justify *de novo* investment is real. But firms should be cautious about projecting major regulatory time savings as a core part of the repurposing value proposition without empirical evidence specific to their particular compound and indication.

From a policy perspective, the findings raise a question about whether current regulatory frameworks adequately capture the efficiency potential of repurposing at the approval stage. If the FDA is effectively treating new indication applications for established compounds with the same thoroughness as applications for genuinely novel compounds, there may be scope for regulatory innovation that recognises the existing knowledge base more explicitly. Breckenridge and Jacob (2019) discuss the potential for revised regulatory pathways that better accommodate repurposing, and the findings of this study provide empirical support for the argument that the current framework does not automatically deliver the time advantages that the repurposing logic would predict.

5.7 Chapter Summary

This chapter has interpreted the empirical findings of the study considering the theoretical framework and the existing literature. Neither H1 nor H2 was confirmed by the data, but the null result is theoretically coherent and contributes meaningfully to the literature rather than simply failing to find an effect.

The most important finding is not about repurposing but about institutional reform. The PDUFA effect dominates the regression results and explains far more of the variation in approval durations during the 1990s than any drug-level characteristic. This finding contextualises the null result for repurposing: the institutional environment was changing so rapidly during this period that drug-level effects were difficult to detect against that backdrop.

The null result for repurposing is consistent with two plausible explanations: either the time advantage of repurposing is concentrated in pre-submission development stages that this study does not measure, or regulatory complexity offsets the theoretical efficiency of repurposing at the approval stage. Both explanations are grounded in the literature, and neither can be ruled out with the available data.

The suggestive pattern in the standard review subgroup, where repurposed drugs showed a 143-day advantage, provides a direction for future research and a theoretically motivated hypothesis about where the repurposing advantage is most likely to be visible. The implications for academic research and industry practice have been discussed, along with an honest account of the study's limitations. The final chapter draws these threads together and offers a concise summary of the study's contributions.

Chapter 6: Conclusion

This final chapter concludes the thesis. It does not introduce new points or present new proofs. Instead, it reviews ideas from earlier chapters, states the research aims and main findings, explains how these results add to existing literature, and suggests directions for future study based on the work presented here. The chapter is deliberately brief as the rigorous reasoning was made before. Here the aim is to give the reader a definite and straightforward description of what this thesis has and has not managed to do, without any pretence.

6.1 Aim of the study

The pharmaceutical industry has been grappling with a productivity crisis that has been both well-documented and longstanding. Over the years, the cost of developing new drugs has escalated dramatically, whereas the number of approvals per dollar spent has decreased. In this context, drug repurposing has been gaining more interest as a potential way to develop drugs more efficiently than the traditional approach, with theoretical studies suggesting it could shorten development time by up to five years and lower costs by up to 85%. These statements

are very frequently quoted and have influenced the academic discussion as well as the pharmaceutical industry's strategy.

Yet despite the confidence with which these efficiency arguments are advanced, direct empirical evidence on one of their most specific and testable implications is surprisingly thin. If repurposing really is faster, this should be observable in regulatory approval data. A drug with an established safety profile and known pharmacological properties should, in principle, move through the FDA approval process more quickly than a genuinely new compound facing the same evidentiary requirements from scratch. This prediction is grounded in both the pharmaceutical repurposing literature and the theoretical framework of exaptation developed by Andriani et al. (2017) and applied to repurposing by Rake and Hanisch (2023; 2024).

This thesis tested that prediction directly. Using data from the FDA Drugs@FDA database, a dataset of 289 Type 1 new molecular entity NDA submissions from 1990 to 1999 was constructed. Approval durations were extracted from original FDA approval letters, and drugs were classified as repurposed or non-repurposed based on evidence of subsequent approval for a new therapeutic indication. Two hypotheses were tested: H1, that repurposed drugs have shorter regulatory approval timelines, and H2, that any observed difference reflects genuine process-level effects rather than compositional differences in the sample.

6.2 Findings

The two hypotheses tested in this study can be summarised as follows. H1, that repurposed drugs are associated with shorter regulatory approval timelines, was not confirmed. The directional pattern supports the hypothesis, but the difference is not statistically significant. H2, that observed differences reflect genuine process-level effects rather than compositional differences, was also not confirmed. The dominant finding is that submission year, reflecting the institutional impact of PDUFA in 1992, explains approval duration variation far more strongly than repurposing status in this sample.

The results present a nuanced and honest picture. At the descriptive level, repurposed drugs were faster. They averaged 538 days from submission to approval compared to 609 days for non-repurposed drugs, a difference of 71 days. The median difference was 39 days. The direction of this pattern is consistent with H1 and with the theoretical predictions of the repurposing literature.

However, none of the statistical tests confirmed this difference as significant. The Mann-Whitney U test returned a two-sided p-value of 0.908. The Chi-squared and Fisher's exact tests both returned p-values of approximately 0.50. The negative binomial regression, which is the appropriate primary model for this type of over dispersed count data, produced an incidence rate ratio of 0.877 for repurposing in the baseline model, falling to 0.952 once submission year and review priority were added as controls. Neither estimate reached conventional significance thresholds. H1 is not confirmed. H2 is not confirmed either. The interaction term testing whether the repurposing effect differed across regulatory pathways was essentially zero.

The dominant finding of the study is not about repurposing at all. It is about institutional reform. Submission year is by far the strongest predictor of approval duration, with each additional year associated with an approximately 18% reduction in expected approval time. This reflects the impact of the Prescription Drug User Fee Act of 1992, which introduced performance-based review targets and fundamentally changed how quickly the FDA processed new applications. The PDUFA effect is large, robust, and consistent across all model specifications. By comparison, repurposing status contributes very little to explaining differences in approval duration once this institutional context is accounted for.

The subgroup analysis produced one finding worth noting. In the standard review subgroup, repurposed drugs averaged 521 days compared to 664 days for non-repurposed drugs, a gap of 143 days. This is suggestive and theoretically interpretable but did not reach statistical significance, partly because of the reduced sample sizes within subgroups. It represents a direction for future research rather than a confirmed finding.

6.3 Contributions

This study makes two contributions to the existing literature. The first is empirical. It provides one of the relatively few systematic quantitative analyses of FDA approval durations comparing repurposed and non-repurposed drugs using a drug-level dataset with precise date-level measurement. The existing literature has advanced the efficiency case for repurposing primarily through theoretical arguments and broad industry estimates. This study subjects one specific and testable implication of those arguments to direct empirical scrutiny and finds that it is not supported in this sample at the regulatory approval stage. This is a useful corrective. It does not disprove the efficiency case for repurposing more broadly, but it does challenge the confidence with which regulatory time savings are claimed as a core advantage.

The second contribution is methodological. The study demonstrates the importance of controlling for institutional context when studying pharmaceutical approval timelines. The PDUFA finding shows that regulatory reforms can produce effects so large that they swamp drug-level differences in observational data. Any study of approval durations that does not account for the timing of major regulatory reforms risks misattributing institutional effects to drug-level characteristics. This is a general lesson that extends beyond the specific question of repurposing.

The study also contributes to the theoretical conversation by identifying where the efficiency gains from repurposing are most likely and least likely to manifest. The null result at the regulatory approval stage, taken together with the broader theoretical framework, suggests that the time savings associated with repurposing are probably concentrated in pre-submission development activities, where safety data eliminates preclinical and Phase I requirements, rather than in the regulatory review itself. This reframing of where in the development pipeline the repurposing advantage operates is a theoretically productive contribution even in the absence of a positive empirical finding.

6.4 Limitations

Extensive discussion on the main limitations of the study has been made in Chapter 5. A summary is provided below. The key challenge relates to missing data. Among 289 submissions in the entire sample, exact submission date records for 140 submissions could not be located, so 148 cases were used for the main analysis. It is hard to determine if this smaller group truly represents the whole population. Also, the sample comprises only FDA Type 1 new molecular entity submissions from the period 1990 to 1999, thereby making the findings hardly applicable to other drug types, regulatory jurisdictions, and periods. Considering the study is observational, it only reveals correlations but not causations. Moreover, since the focus of the study is on the regulatory approval phase, it remains unknown whether repurposing would lead to time savings at the earlier development stages which, from a theoretical viewpoint, is where the greatest benefit would be derived.

6.5 Directions for Future Research

Several directions of further research can be considered based on the points and limitations of this study. The most basic extension is to revisit the study procedure with a bigger database spanning more up-to-date approval periods and including a full set of submission dates. A bigger sample would increase the statistical power to identify moderate effects and would

enable more meaningful subgroup analyses, particularly the comparison between standard and priority review drugs that yielded the most suggestive pattern in this case study.

A more challenging extension of this work might be to measure the whole drug development timeline instead of just the regulatory approval phase. In case registration of trial initiation dates was done besides submission and approval dates, analysing the overall development time from the first clinical trial to market for repurposed and non-repurposed drugs could be done. Therefore, it would be possible to directly verify whether the time saved by repurposing drugs is only at the pre-submission phase, as the theoretical model assumes, or also includes the regulatory phase in ways that a bigger sample might disclose.

Furthermore, repurposing research could be advanced to identify whether the link between the drug repurposing and drug approval time is dependent on the therapeutic area, the type of diseases, or the degree of similarity between the original and new drug indications. The concept of a drug exaptation introduced by Mastrogiorgio and Gilsing (2016) stress that the ability of a drug to be used for new purposes might be related to its technological complexity. Drugs with more complex mechanisms of action or broader pharmacological profiles may demonstrate more significant drug repurposing efficiency. Addressing this question would bring this empirical research closer to the theoretical forecasts.

Lastly, the policy implication outlined in Chapter 5 that current regulatory frameworks may not be optimized to fully realize the efficiency gains of repurposing is something that the research community involved in innovation policy and pharmaceutical regulation should be concentrating on. A comparative study of approval times in various regulatory jurisdictions with different frameworks for the handling of repurposed drugs might indicate whether alternative regulatory models result in systematically different outcomes.

6.6 Closing Remarks

Drug repurposing is indeed an innovative and important strategy for drug development. Its efficiency advantages are strongly argued theoretically; its role in the pharmaceutical industry is large; and its capability to meet the needs of healthcare systems for drugs from less profitable markets cannot be ignored. None of this is contested. What this paper has done is only one aspect of efficiency, regulatory approval time, which was tested with actual data and correct methods, and it was found that the evidence does not support it at the usual significance levels for this particular sample.

It does not discourage us from looking for more evidence. A negative outcome that is in line with theory, has a good methodology, and is honestly reported can be a valid and useful knowledge contribution. Drug repurposing literature has generally accepted time efficiency for granted rather than questioning. This work regards it as a question and finds that the evidence is not in favour of the regulatory confirmation of that. The difference here is significant for directing repurposing strategies by firms, for setting regulatory frameworks by policymakers, and for conducting further empirical research by scholars.

The biggest result out of the whole study is the least complicated one. The most important determinant of how quickly a drug received FDA approval in the 1990s was not the nature of the drug itself but when it was submitted. Institutional settings are extremely important. The Prescription Drug User Fee Act altered the regulatory environment so much that its impacts are greater than those for any drug characteristic found in the data. Knowing the institutional side of pharmaceutical innovation is not just a story's background. In this paper at least, it is the story.

References

- Andriani, P. and Kaminska, R. (2021) 'Exploring the dynamics of novelty production through exaptation: a historical analysis of coal tar-based innovations', *Research Policy*, 50(2), p. 104171. <https://doi.org/10.1016/j.respol.2020.104171>
- Andriani, P., Ali, A. and Mastrogiorgio, M. (2017) 'Measuring exaptation and its impact on innovation, search, and problem solving', *Organization Science*, 28(2), pp. 320–338. <https://doi.org/10.1287/orsc.2017.1116>
- Ashburn, T.T. and Thor, K.B. (2004) 'Drug repositioning: identifying and developing new uses for existing drugs', *Nature Reviews Drug Discovery*, 3(8), pp. 673–683. <https://doi.org/10.1038/nrd1468>
- Baker, T. and Nelson, R.E. (2005) 'Creating something from nothing: resource construction through entrepreneurial bricolage', *Administrative Science Quarterly*, 50(3), pp. 329–366. <https://doi.org/10.2189/asqu.2005.50.3.329> [Cited in Rake and Hanisch, 2023].
- Breckenridge, A. and Jacob, R. (2019) 'Overcoming the legal and regulatory barriers to drug repurposing', *Nature Reviews Drug Discovery*, 18(1), pp. 1–2. <https://doi.org/10.1038/nrd.2018.92>
- Cameron, A.C. and Trivedi, P.K. (2013) *Regression Analysis of Count Data*. 2nd edn. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9781139013567>
- Chesbrough, H.W. (2006) 'Open innovation: a new paradigm for understanding industrial innovation', in Chesbrough, H.W., Vanhaverbeke, W. and West, J. (eds.) *Open Innovation: Researching a New Paradigm*. Oxford: Oxford University Press, pp. 1–12. <https://doi.org/10.1093/oso/9780199226467.001.0001>
- DiMasi, J.A., Grabowski, H.G. and Hansen, R.W. (2016) 'Innovation in the pharmaceutical industry: new estimates of R&D costs', *Journal of Health Economics*, 47, pp. 20–33. <https://doi.org/10.1016/j.jhealeco.2016.01.012>
- Field, A. (2013) *Discovering Statistics Using IBM SPSS Statistics*. 4th edn. London: SAGE Publications. <https://uk.sagepub.com/en-gb/eur/discovering-statistics-using-ibm-spss-statistics/book257672>
- Fox, J. (2016) *Applied Regression Analysis and Generalized Linear Models*. 3rd edition. London: SAGE Publications. <https://doi.org/10.4135/9781452230572>
- Hanisch, M. and Rake, B. (2021) 'Repurposing without purpose? Early innovation responses to the COVID-19 crisis: evidence from clinical trials', *R&D Management*, 51(4), pp. 393–409. <https://doi.org/10.1111/radm.12466> [Cited in Rake and Hanisch, 2023].

- Hay, M., Thomas, D.W., Craighead, J.L., Economides, C. and Rosenthal, J. (2014) 'Clinical development success rates for investigational drugs', *Nature Biotechnology*, 32(1), pp. 40–51. <https://doi.org/10.1038/nbt.2786>
- Kesselheim, A.S., Tan, Y.T. and Avorn, J. (2015) 'The roles of academia, rare diseases, and repurposing in the development of the most transformative drugs', *Health Affairs*, 34(2), pp. 286–293. <https://doi.org/10.1377/hlthaff.2014.1038> [Cited in Rake and Hanisch, 2023].
- Long, J.S. and Ervin, L.H. (2000) 'Using heteroscedasticity consistent standard errors in the linear regression model', *The American Statistician*, 54(3), pp. 217–224. <https://doi.org/10.1080/00031305.2000.10474549>
- Mastrogiorgio, M. and Gilsing, V. (2016) 'Innovation through exaptation and its determinants: the role of technological complexity, analogy making and patent scope', *Research Policy*, 45(7), pp. 1419–1435. <https://doi.org/10.1016/j.respol.2016.04.003>
- Parvathaneni, V., Kulkarni, N.S., Muth, A. and Gupta, V. (2019) 'Drug repurposing: a promising tool to accelerate the drug discovery process', *Drug Discovery Today*, 24(10), pp. 2076–2085. <https://doi.org/10.1016/j.drudis.2019.06.014>
- Paul, S.M., Mytelka, D.S., Dunwiddie, C.T., Persinger, C.C., Munos, B.H., Lindborg, S.R. and Schacht, A.L. (2010) 'How to improve R&D productivity: the pharmaceutical industry's grand challenge', *Nature Reviews Drug Discovery*, 9(3), pp. 203–214. <https://doi.org/10.1038/nrd3078>
- Pushpakom, S., Iorio, F., Eyers, P.A., Escott, K.J., Hopper, S., Wells, A., Doig, A., Williams, T., Latimer, J., McNamee, C., Norris, A., Sanseau, P., Cavalla, D. and Pirmohamed, M. (2019) 'Drug repurposing: progress, challenges and recommendations', *Nature Reviews Drug Discovery*, 18(1), pp. 41–58. <https://doi.org/10.1038/nrd.2018.168>
- Rake, B. and Hanisch, M. (2023) 'Repurposing: a collaborative innovation strategy for the digital age', in Bstieler, L. and Noble, C.H. (eds.) *The PDMA Handbook of Innovation and New Product Development*. 4th edn. Hoboken: John Wiley and Sons, pp. 103–119. <https://doi.org/10.1002/9781119691983.ch7>
- Rake, B. and Hanisch, M. (2024) 'Innovation repurposing', in *Encyclopedia of Creativity, Invention, Innovation and Entrepreneurship*. Amsterdam: Elsevier. <https://doi.org/10.1007/978-3-319-15347-6>
- Scannell, J.W., Blanckley, A., Boldon, H. and Warrington, B. (2012) 'Diagnosing the decline in pharmaceutical R&D efficiency', *Nature Reviews Drug Discovery*, 11(3), pp. 191–200. <https://doi.org/10.1038/nrd3681>
- Wooldridge, J.M. (2010) *Econometric Analysis of Cross Section and Panel Data*. 2nd edition. Cambridge, MA: MIT Press. <https://mitpress.mit.edu/9780262232586>

Appendix A: Variable Definitions and Coding

Table A1 provides a complete description of all variables used in the regression analysis, including how each variable was operationalised, its coding scheme, the values it takes in the analysis dataset, and the source from which it was derived.

Table A1: Variable Definitions, Coding, and Sources

Variable	Role in model	Coding	Values in sample	Source
Approval Duration	Dependent variable	Days from first NDA submission date to FDA approval date	62 to 3,502 days. Mean 583d, median 415d (N=148)	Extracted from original FDA approval letters via Drugs@FDA database
Log (Approval Duration)	Dependent variable-OLS robustness checks only	Natural log of approval duration in days. Applied to reduce right skew for OLS regression	4.13 to 8.16 (log scale)	Derived from Approval Duration variable
Repurposed	Primary independent variable	Binary: 1 = drug subsequently received FDA approval for a new therapeutic indication; 0 = no such approval found	1 = Repurposed: N=122 (42.2%). 0 = Non-repurposed: N=167 (57.8%)	Classified by reviewing FDA regulatory history and drug labels for each application. Decision rule described in Section 3.4.3
Priority Review	Control variable	Binary: 1 = FDA priority review designation; 0 = standard review designation	1 = Priority: N=122. 0 = Standard: N=166. Missing: N=1	FDA Drugs@FDA database, ReviewPriority field
Submission Year	Control variable-temporal trend / PDUFA effect	Continuous integer: calendar year of first NDA submission	1990 to 1999. Ranges from N=18 in 1994 to N=52 in 1996	FDA Drugs@FDA database, Submission_Year field
Discontinued	Control variable	Binary: 1 = drug subsequently discontinued from market; 0 = prescription or	1 = Discontinued: N=180 (62.3%). 0 = Other: N=109 (37.7%)	FDA Drugs@FDA database, MarketingStatusDescription field. Discontinuation is post-approval and does not

Variable	Role in model	Coding	Values in sample	Source
		OTC status retained		affect approval duration measure
Marketing Status	Descriptive-full 3-category version used for subgroup analysis	Categorical: Prescription / Discontinued / Over the counter (OTC)	Prescription: N=105. Discontinued: N=180. OTC: N=4	FDA Drugs@FDA database MarketingStatusDescription field
Repurposed × Priority	Interaction term- Model 4 only (H2 test)	Product of Repurposed and Priority binary variables. Value = 1 only when drug is both repurposed and priority-reviewed	1 = Repurposed AND Priority: N=23. 0 = all other combinations: N=125	Derived from Repurposed and Priority variables. Included in Model 4 to test whether repurposing effect differs across regulatory pathways (H2)

Note: All variables were constructed from the FDA Drugs@FDA database using the 289-application full sample. The regression analysis used the 148-observation subset with complete approval duration data. Binary variables are coded 0/1. The Repurposed × Priority interaction term appears in Model 4 only and is not part of the primary three-model specification.

Appendix B: Comparison of Complete and Incomplete Observations

As noted in Chapter 3, 140 of the 289 applications did not have recoverable exact submission dates, reducing the analysis sample to 148 observations. Table B1 compares the composition of the complete and incomplete subsets across key characteristics to assess whether the missing data is likely to have introduced systematic bias into the results.

Table B1: Comparison of Complete vs Incomplete Observations

Characteristic	Complete (N=148)	Incomplete (N=141)	Interpretation
% Repurposed	37.2% (N=55)	47.5% (N=67)	Incomplete subset contains slightly more repurposed drugs. This means the analysis sample modestly underrepresents repurposed drugs, which would bias against finding a significant repurposing effect, consistent with the null result
% Priority Review	41.2% (N=61)	43.3% (N=61)	Very similar proportions. Priority review composition is unlikely to differ meaningfully between the two subsets
% Discontinued	60.1% (N=89)	64.5% (N=91)	Similar proportions. Discontinued status does not appear systematically related to data availability
Mean submission year	1995.4	1994.8	Incomplete observations skew slightly earlier, consistent with older records being harder to retrieve from the FDA database. The small difference of 0.6 years is unlikely to materially affect the year coefficient in the regression

Note: Incomplete observations include the 140 applications with missing submission dates and the one application excluded for a negative approval duration value. The slight overrepresentation of repurposed drugs in the incomplete subset (47.5% vs 37.2%) means any resulting bias works against detecting a significant repurposing effect, which is consistent with the null result reported in Chapter 4.

Appendix C: FDA Approval Letter

NDA 20-835

MAR 27 1998

Procter and Gamble Pharmaceuticals
Attention: Dr. Hina Wu
Senior Scientist, Regulatory Affairs
11450 Grooms Road
Cincinnati, OH 45242

APPEARS THIS WAY
ON ORIGINAL

Dear Dr. Wu:

Please refer to your new drug application dated March 31, 1997, received April 1, 1997, submitted under section 505(b) of the Federal Food, Drug, and Cosmetic Act for Actonel (risedronate sodium) 30 mg Tablets.

We acknowledge receipt of your submissions dated April 16, July 25 and 31, September 26, October 14 and 16, and December 5, 10, 12, 18, and 19, 1997; and January 6, 7, and 16, February 10, 13, 18, 20, 23, and 24, and March 2, 4, and 12, 1998. The User Fee goal date for this application is April 1, 1998.

This new drug application provides for the treatment of Paget's disease of bone.

We have completed the review of this application, including the submitted draft labeling, and have concluded that adequate information has been presented to demonstrate that the drug product is safe and effective for use as recommended in the draft labeling in the submission dated March 12, 1998. Accordingly, the application is approved effective on the date of this letter.

The final printed labeling (FPL) must be identical to the draft patient package insert submitted on March 12, 1998, and draft packaging labeling dated March 10, 1998. Marketing the product with FPL that is not identical to this draft labeling may render the product misbranded and an unapproved new drug.

Please submit 20 copies of the FPL as soon as it is available, in no case more than 30 days after it is printed. Please individually mount ten of the copies on heavy-weight paper or similar material. For administrative purposes, this submission should be designated "FINAL PRINTED LABELING for approved NDA 20-835." Approval of this submission by FDA is not required before the labeling is used.

Should additional information relating to the safety and effectiveness of the drug become available, revision of that labeling may be required.

In addition, please submit three copies of the introductory promotional material that you propose to use for this product. All proposed materials should be submitted in draft or mock-up form, not final print. Please submit one copy to the Division of Metabolic and Endocrine Drug Products and two copies of both the promotional material and the package insert directly to:

**APPEARS THIS WAY
ON ORIGINAL**

Food and Drug Administration
Division of Drug Marketing, Advertising and Communications,
HFD-40
5600 Fishers Lane
Rockville, Maryland 20857

Validation of the regulatory methods has not been completed. At the present time, it is the policy of the Center not to withhold approval because the methods are being validated. Nevertheless, we expect your continued cooperation to resolve any problems that may be identified.

Please submit one market package of the drug product when it is available.

We remind you that you must comply with the requirements for an approved NDA set forth under 21 CFR 314.80 and 314.81.

If you have any questions, please contact Randy Hedin, R.Ph., Senior Regulatory Management Officer, at (301)827-6392.

Sincerely yours,

**APPEARS THIS WAY
ON ORIGINAL**

James Bilstad, M.D.
Director
Office of Drug Evaluation II
Center for Drug Evaluation and Research

**APPEARS THIS WAY
ON ORIGINAL**

cc:

Original NDA 20-835
HFD-510/Div. files
HFD-510/CSO/R.Hedin
HFD-510/GTroendle/GKuijpers/DColeman/RSteigerwalt/CJones/HAhn